

Recombination of binding domains shapes structural evolution of *Betacoronavirus* spike proteins

Neil Scheidwasser¹✉, Mark P. Khurana¹, Christian Morgenstern², Nina Pust¹, Prince Ravi Leow¹, Rachel Kavanagh¹, Samir Bhatt^{1,2†} & David Duchêne^{1†}

¹Section of Health Data Science & AI, Department of Public Health, University of Copenhagen, Copenhagen, Denmark; ²MRC Centre for Global Infectious Disease Analysis, Imperial College London, London, United Kingdom; [†]Equal contribution

Correspondence: neil.clow@sund.ku.dk

The spike protein of betacoronaviruses mediates host cell entry and exhibits remarkable receptor usage flexibility. While recombination has been identified as a powerful evolutionary force in *Betacoronavirus* genomes, its specific contribution to the structural changes that enable receptor switching remains unclear. Here, we employed phylogenetic multi-tree mixture modelling to analyse experimentally determined spike protein structures and their corresponding 3Di structural representations. Our results refine previous models of *Betacoronavirus* protein evolution, suggesting that while recombination events and gradual mutations in the N-terminal domain significantly drive ACE2 binding capability at the sequence level, host tropism seems mediated by gradual structural refinement within the receptor-binding domain rather than by large-scale structural reorganisation. Our findings highlight the necessity of integrating sequence- and structure-based evolutionary analyses to understand viral host adaptation, as sequence-level diversification does not necessarily translate into the structural determinants of host range, with important implications for zoonotic risk prediction and intervention design.

Betacoronaviruses constitute a significant threat to global health, causing three major epidemics in the past two decades — SARS [1–3], MERS [4], and COVID-19 [5–7] — alongside other recurring outbreaks of upper respiratory infections [8, 9]. With ecosystems under mounting stress from deforestation and climate change, spillover events from bat and rodent reservoirs [10–12] are likely to accelerate the emergence of novel strains by exerting strong evolutionary selection pressures [13, 14]. One salient feature of *Betacoronavirus* emergence is the flexibility in host receptor usage, enabling infection across diverse groups of mammals [15–18] which may facilitate zoonotic transmission [19, 20]. This flexibility mainly stems from innovations in the viral spike glycoprotein, which mediates attachment and membrane fusion with host cells [5, 21]. These mechanisms are mediated by two key subunits, with receptors for cell-binding in S1 and cofactors for membrane fusion in S2 [20–23]. Antibody escape and improved host affinity occur via critical mutations in the spike gene [24, 25], as well as recombination across variants in coinfection scenarios [26, 27]. The spike gene is thus central to *Betacoronavirus* novelty and has been proposed to play a role in host switching and the onset of pathogenic outbreaks in humans [27–30]. However, the primary molecular evolutionary processes driving novelty remain unclear [27]. Recombination is considered a powerful evolutionary force [31, 32], yet its influence on spike protein structure has been suggested to be limited [30, 33, 34], raising questions regarding its role in the emergence of high-risk strains.

Evolutionary analyses of *Betacoronavirus* emergence have traditionally been sequence-based. In the wake of the COVID-19 pandemic, large-scale genomic surveillance has yielded unprecedented insights into coronavirus transmission dynamics and evolutionary landscapes [32, 35–37].

Meanwhile, recent breakthroughs in protein structure prediction [38–40] have catalysed studies investigating protein evolution through a structural lens [41–48], since protein structures directly underpin biological function [49] and are natural targets of evolutionary selection. In the case of spike proteins, the receptor binding domain (RBD), typically found in S1's C-terminal region [22], contains multiple sites for host binding [50, 51], suggesting that structural change is of evolutionary importance. However, major structural changes were not found to be central to the spread of the Omicron SARS-CoV-2 variant [52], suggesting an important role for subtle structural evolutionary changes. The N-terminal domain (NTD) of the S1 subunit has multiple binding sites and may vary greatly in transmission efficiency after recombination events [53]. Deep learning-based alphabet representations of protein structure such as 3Di [54] offer a window into the relative roles of mutation and recombination in driving structural change in betacoronaviruses, but this is only possible via advanced molecular evolutionary modelling.

Here, we leverage experimentally derived structures of spike glycoproteins from various *Betacoronavirus* species and apply phylogenetic multi-tree mixture models [61, 67] to amino acid- and structure-derived alignments to dissect the roles of recombination and gradual adaptation in the ability of spike to switch between different host receptors, also known as receptor switching. In coronavirus spike glycoproteins, mixture modelling allows each site in alignments to be probabilistically associated with distinct evolutionary regimes, accommodating multiple evolutionary histories, and thereby capturing the mosaic signal expected under recombination. Our results support a refined model of the evolution of receptor usage. Although amino acid recombination in spike proteins and gradual changes in the S1 NTD

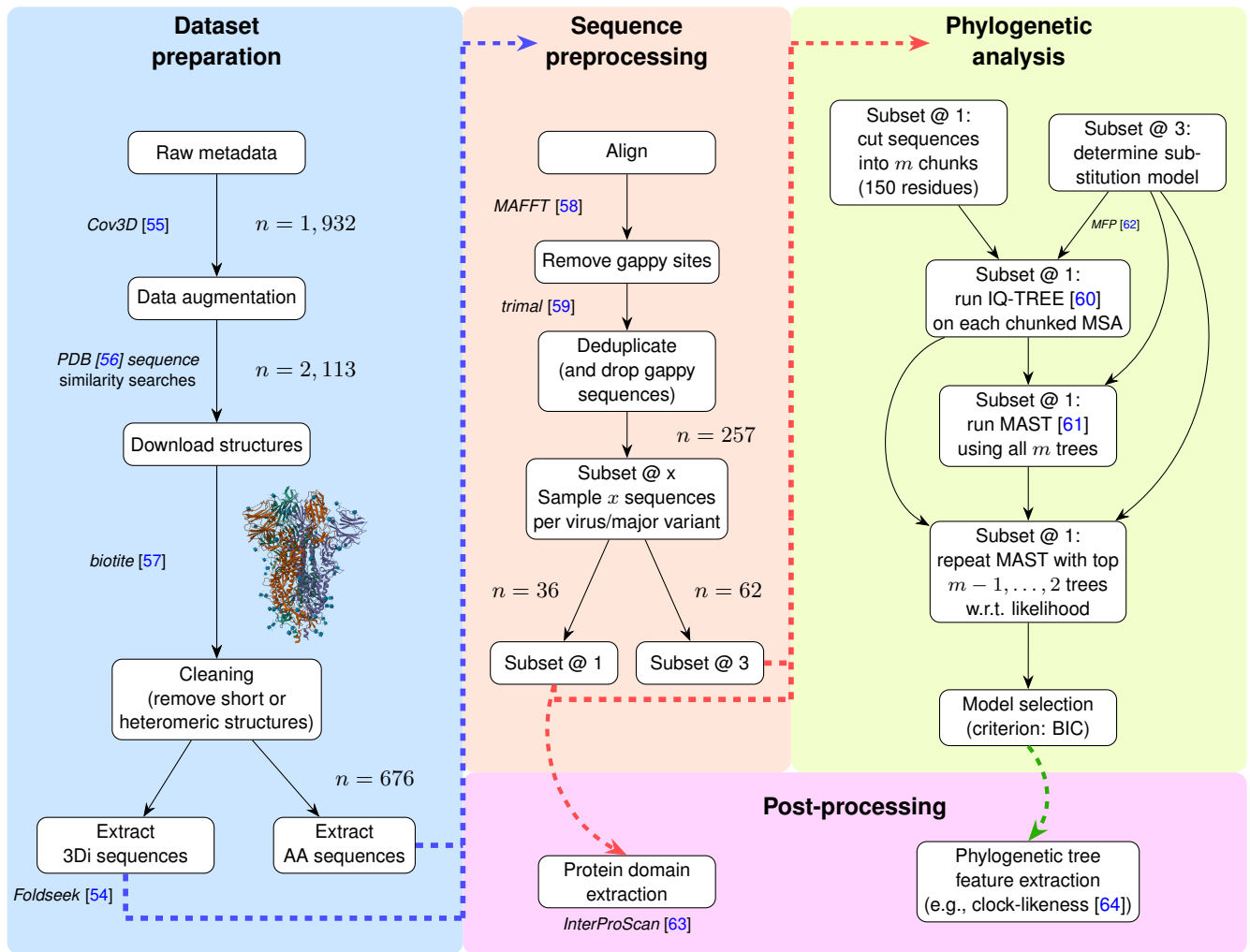


Figure 1. Data processing and analysis workflow for the analysis of experimental spike glycoprotein structures from betacoronaviruses. The molecular structure image of PDB ID 6VXX [65] was obtained from the RCSB PDB (<https://rcsb.org>) using Mol* [66] depicts a spike protein of a wild type SARS-CoV-2 bound with 2-acetamido-2-deoxy-beta-D-glucopyranose (NAG). MFP: ModelFinder Plus [62]. BIC: Bayesian Information Criterion.

both contribute significantly to host adaptation, these processes primarily diversify the sequence background rather than directly altering receptor-binding conformation. From a structural perspective, receptor switching is mediated by small, gradual structural changes within the RBD, rather than large structural shifts driven by recombination or S1 NTD evolution. Thus, although recombination leaves a detectable historical signal in shaping RBD evolution in *Betacoronavirus*, it does not directly produce major structural changes associated with receptor usage.

Multi-tree mixture modelling of spike glycoprotein evolution

Although the advent of protein structure prediction models has propelled the construction of large viral structure datasets [68, 69], they tend to lack the resolution to capture subtle structural differences between variants which

vary by a single or a few mutations [70–73]. Consequently, we curated a diverse dataset of 2,113 *Betacoronavirus* spike protein structures (Fig. 1; Table 1) by augmenting Cov3D database [55] with corrections and sequence similarity searches (see Methods). The resulting dataset spanned 32 *Betacoronavirus* strains including four major variants of SARS-CoV-2 (Alpha, Delta, Omicron, wild-type), from which we built two multiple sequence alignments (MSAs) for amino acid sequences and protein structures mapped to discrete states akin to characters [54] (Fig. 1). For each strain, we selected one representative sample, prioritising high-resolution structures to maximise signal quality and comparison fairness. The alignments span three of the five proposed *Betacoronavirus* subgenera [74] (*Embecovirus*, *Merbecovirus*, *Sarbecovirus*; Fig. 2a), and include viruses that use four major host receptors, the majority of which bind to angiotensin-converting enzyme 2 (ACE2) (Fig. 2b, Table 1).

Both amino acid- and structure-based alignments exhibited

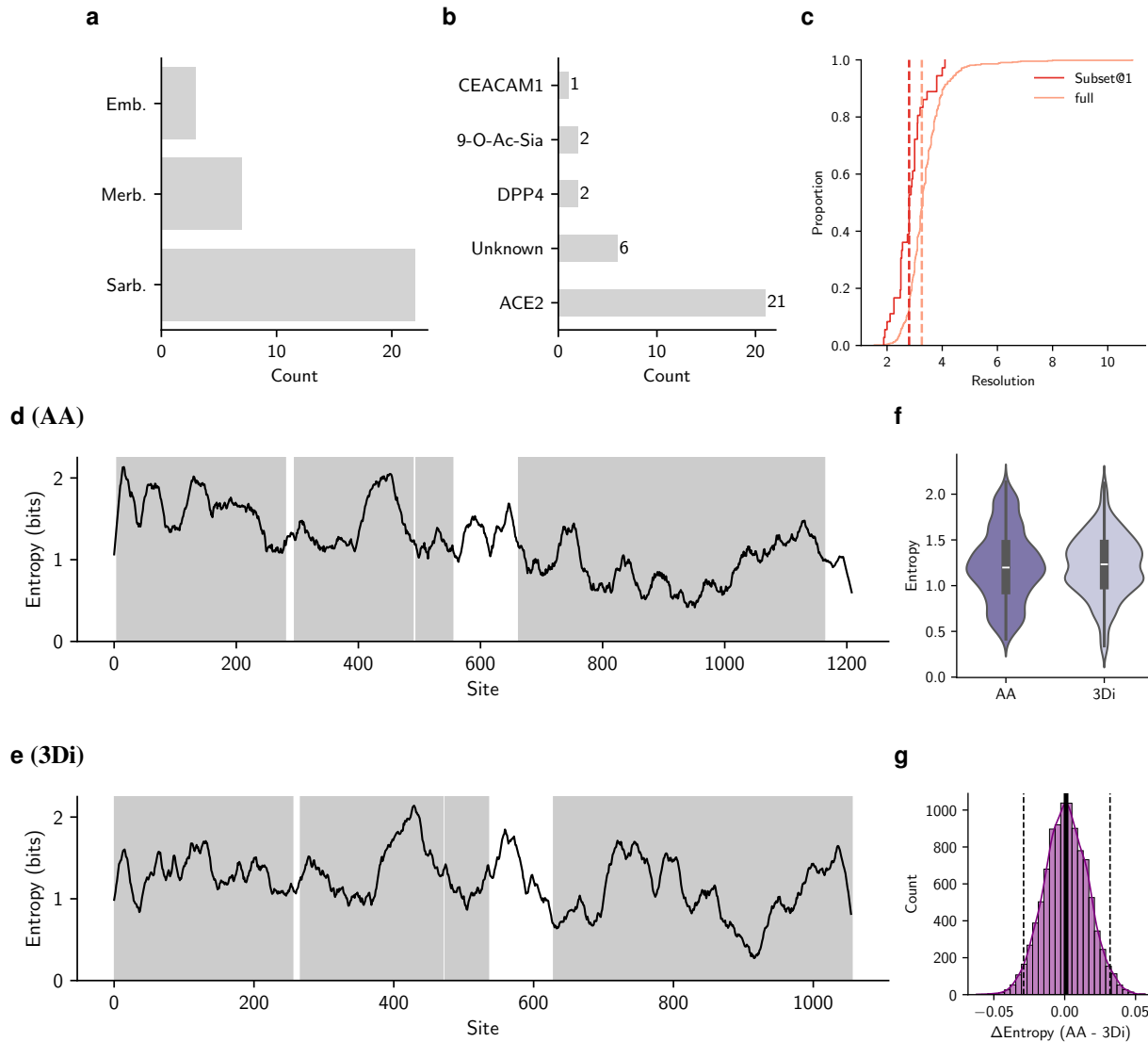


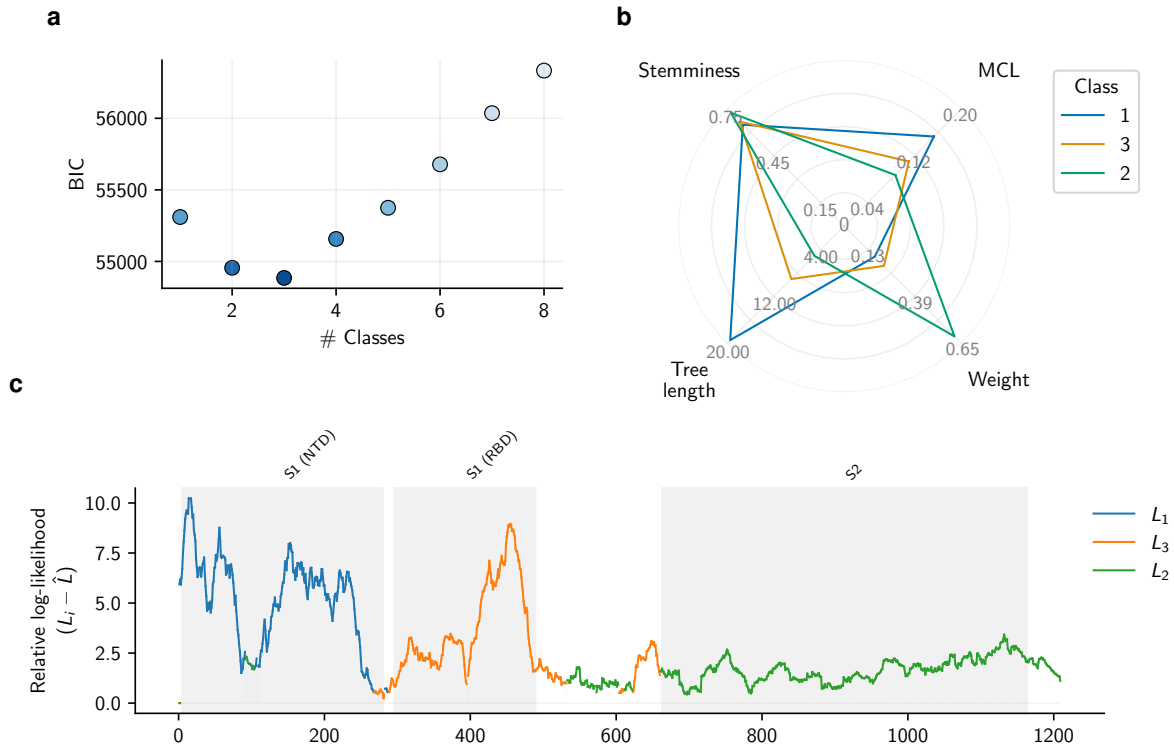
Figure 2. Overview of the amino acid (AA) and 3Di-based alignments of 36 *Betacoronavirus* spike glycoproteins. **(a)** *Betacoronavirus* subgenus distribution in the dataset. Sarb.: *Sarbecovirus*; Merb.: *Merbecovirus*; Emb.: *Embecovirus*. **(b)** Receptor distribution. CAECAM1: Carcinoembryonic antigen-related cell adhesion molecule 1. 9-O-Ac-Sia: 9-O-acetylated sialic acid. DPP4: Dipeptidyl peptidase-4 inhibitor. ACE2: angiotensin-converting enzyme 2. **(c)** Distribution of protein structure resolution in the full dataset (2,113 structures) and in the final dataset (‘Subset@1’; 36 structures) used for downstream phylogenetic analyses. **(d, e)** Site-wise Shannon entropy of the amino acid ((d)) and structural ((e)) alignments. Shaded areas represent three domains of the S1 subunit (NTD, RBD, CTD), and the S2 subunit, inferred via InterProScan [63]. **(f)** Distribution of site-wise entropy in (d) and (e). **(g)** Bootstrapped differences in site-wise entropy between the AA and 3Di dataset. Dashed lines denote the lower and upper bounds of the bootstrapped 95% confidence interval, whereas the thicker line represents the mean.

similar and expected patterns across key regions of the spike protein, such as higher conservation in the S2 subunit and higher diversity in the S1 RBD [51, 75, 76] (Fig. 2d, 2e). Surprisingly, the structural alignment was not significantly more conserved than the structural MSA in terms of site entropy (mean difference: 0.001 [95% bootstrapped confidence interval (CI): -0.029 — 0.032]; Fig. 2f, 2g). Although protein structures are known to be noticeably more conserved than sequences [77], this may reflect discretization artifacts inherent to the 3Di structural alphabet, such as

its tendency to favour a few states [43] and sensitivity to minor conformational variations near state boundaries.

Using phylogenetic mixture modelling across sites and trees (MAST) [61], we recovered distinct evolutionary patterns across the spike protein (Fig. 3), suggesting recombination of gene subregions, notably the receptor-binding domain (RBD). Importantly, we found support for three recombinant classes in both amino acid and structural alignments (Fig. 3a, Fig. 3d) which cover similar regions across the spike gene,

Amino acid MSA



3Di MSA

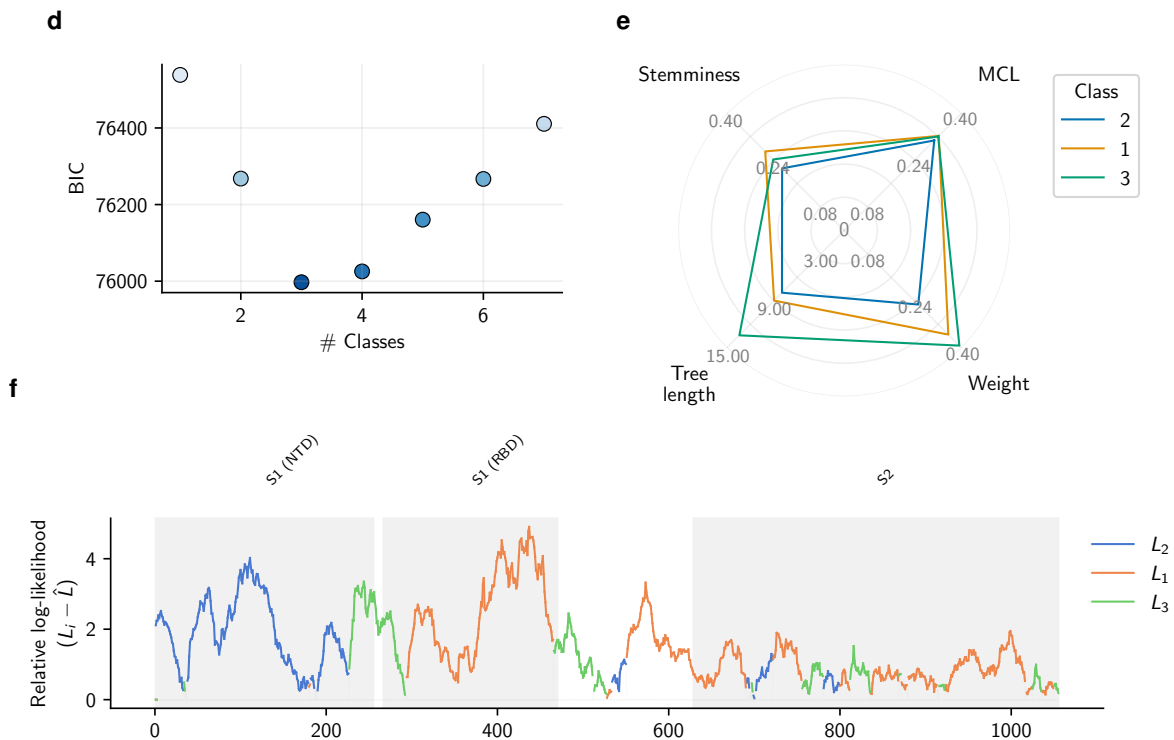


Figure 3. Phylogenetic mixture modelling of recombination in the spike protein of coronaviruses. **(a), (d)** Bayesian information criteria (BIC) of MAST models against different numbers of classes. **(b), (e)** Phylogenetic tree statistics for the optimal number of classes determined by model selection. MCL = Molecular clock-likeness [64, 78]; weight = class weight found by MAST, which shares some similarity to gene tree frequency [61]. **(c), (f)** Site-wise likelihoods of each tree of the optimal model. For each given site, a relative likelihood is shown, displaying the difference between the tree with the highest likelihood for that site and the mean of the two others. Shaded areas represent domains inferred using InterProScan on the AA alignment, converted to 3Di coordinates in **(f)**.

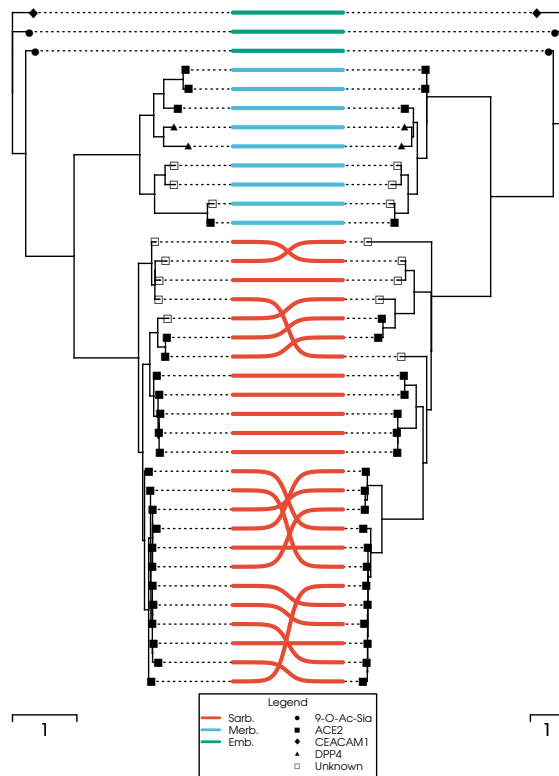
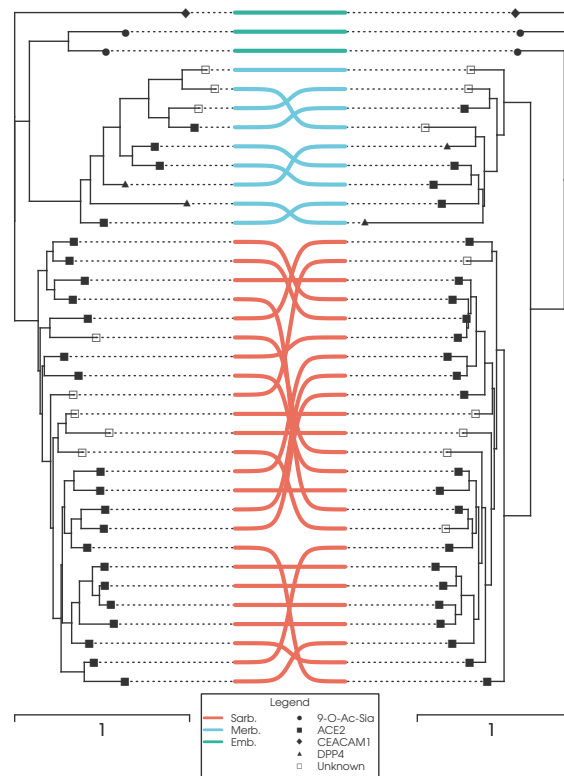
a AA**b 3Di**

Figure 4. Co-phylogenetic plots of trees inferred using MAST [61] covering the S1 NTD (left) and RBD (right) in the AA (a) and 3Di (b) datasets.

notably the N-terminal and receptor-binding domains in the S1 subunit (Fig. 3c, Fig. 3f). We hereinafter refer to the trees corresponding to the class overlapping the RBD and NTD domains in both MSAs as the "RBD tree" and "NTD tree", respectively. In this framework, each class represents a distinct evolutionary regime grouping sites with similar histories, and the resulting trees capture how recombination and adaptation have differently shaped the RBD and NTD.

The NTD tree shows fast evolution in the AA dataset, as illustrated by its high tree length (Fig. 3b, Fig. 3f), but not in the structural dataset (Fig. 3e, Fig. 3f), indicating that rapid sequence divergence does not necessarily correspond to structural change. This pattern is consistent with NTD's role as a secondary receptor-binding domain that, while important for viral attachment in some coronaviruses (e.g., murine coronaviruses, including MCoV A59 [79–81]), may tolerate greater sequence variation without compromising essential functions [82–84].

Conversely, trees overlapping the RBD displayed more complex evolutionary dynamics, consistent with its critical role in determining host range and receptor specificity for betacoronaviruses [82, 85]. The distinct phylogenetic signal observed for RBD-associated sites suggests that this domain experiences unique selective pressures related to receptor-

binding efficiency and immune evasion. In contrast to the NTD and RBD in the S1 subunit, less contiguity was observed between the AA and structural sets for the S2 domain, which is responsible for membrane fusion [17, 86] (Fig. 3c, 3f). While the S2 subunit is primarily represented by a distinct third tree in the amino acid dataset (Fig. 3c), the structural dataset displayed greater ambiguity in class assignments for that region (Fig. 3f). For both datasets, this third tree also had the highest class weight (Fig. 3b, 3e), indicating that it represents the predominant topology across the alignment [61]. This reflects the many constant sites in the AA dataset, whereas its long tree length in the structural dataset can be attributed to the fast evolution of structural sites, nearly approaching saturation (i.e., multiple substitutions accumulating at the same sites). This observation is also plausible from a structural perspective, as the heptad regions in the S2 domain have a high alpha helical content, which has been shown to lead to poor performance on 3Di MSAs [43].

Taken together with Fig. 2, our results showed more conserved evolutionary patterns for S2-overlapping classes, reflecting stringent functional constraints on fusion machinery. This apparent conservation is in agreement with previous research [86, 87], since S2 must maintain precise structural conformations to mediate the complex process of viral-host

membrane fusion, leaving little tolerance for disruptive mutations [30].

Molecular and structural determinants of ACE2 binding in *Betacoronavirus*

To investigate the evolutionary mechanisms underlying ACE2 binding, we compared the molecular and structural trees covering the RBD, whereby a receptor-binding motif is directly responsible for interacting with ACE2 [21, 25, 51, 75, 88], and trees covering the NTD, whose evolution is broadly affected by gradual mutations [30, 82]. Co-phylogenetic plots of NTD and RBD trees (Fig. 4a, 4b) revealed consistent clustering of *Betacoronavirus* subgenera for both subsets, although heightened discordance was observed in the structural trees. However, consistent with recombination shaping domain-specific evolutionary histories, we observed several topological discordances between RBD and NTD trees in both datasets, particularly for ACE2-binding taxa. Using these phylogenies, we aimed to quantify the contributions of recombination and mutation to ACE2 binding. Recombination was measured using a custom taxon discordance index (TDI) which captures each taxon's impact on the mutual clustering information of two trees [89], while mutations were represented by root-to-tip distances from the RBD and NTD trees. Using Bayesian regression modelling [90], we observed that root-to-tip length in NTD trees (3.41 [95% credible interval (CrI): 1.48 – 6.17] and the TDI across both trees (-1.85 [95% CrI: -3.61 – -0.40]) were significant predictors of ACE2 binding (Fig. 5b), indicating *a priori* that gradual adaptation in the S1 NTD and recombination in the S1 subunit significantly mediates ACE2 binding. These observations align with prior studies showing that recombination within the S1 subunit can generate spike variants with altered receptor usage [20, 91, 92]. In addition, NTD is a known recombination "cold spot" [30, 82], and gradual adaptation in the NTD has been proposed to mediate RBD conformation and accessibility for ACE2 usage [93, 94].

In the structural dataset, only the root-to-tip length in the *structural* RBD tree (-7.98 [95% CrI: -17.36 – -0.97]) contributed significantly to explaining a taxon's propensity to bind to ACE2. Taken together, this suggests that recombination events in the S1 subunit and gradual adaptation in its NTD domain only play an upstream role in receptor usage, but only translate into major shifts in protein structure. Indeed, insights from the structural dataset show that alterations in receptor usage are primarily driven by subtle structural changes within the RBD, rather than gradual modifications in the NTD or large-scale recombination events. Consequently, these results supports scenarios where intermediate hosts facilitate stepwise adaptation to human receptors [25, 27, 82, 95]. Although recombination plays a role

in shaping the receptor-binding domain of *Betacoronavirus* spike proteins [30, 91], it may not translate into meaningful changes in protein structure, as subtle modifications of the protein structure seem to primarily drive ACE2 affinity [96].

Discussion

Our analysis provides new insights into the evolution of *Betacoronavirus* spike glycoproteins. Yet, several limitations should be acknowledged. Although simple and accessible, the 3Di violates the assumption of site-wise statistical independence that underpins likelihood-based phylogenetic inference [46, 47]. Alternatively, structural convergence at functionally important sites may cause some viral strains to appear more similar than they are, thereby obscuring the true phylogenetic relationships among the subgenera of interest. Furthermore, previous studies have disputed the reliability of protein structure-derived trees for resolving recent divergences [42], which is the case in coronavirus evolution [97, 98]. However, mixture modelling approaches, such as MAST [61], used here, relax the traditional assumption that a single tree may accurately describe an alignment, allowing the modelling of recombination processes, as observed in this study. Although [48] suggested the use of edge-unlinked models, we argue that this might create an over-parametrised model which may be difficult to fit to our datasets. Finally, our focus on spike proteins, while justified by their critical role in the viral replication and host adaptation of betacoronaviruses, remains an incomplete picture of their evolutionary dynamics. Therefore, future studies should expand beyond spike proteins to investigate the structural evolution of other critical viral components. Among others, a host of proteins, such as the nucleocapsid protein [99, 100] as well as many non-structural proteins, have been found to influence immune evasion [101] and constitute a target for vaccine strategies [102].

In conclusion, phylogenetic mixture modelling analysis reveals that recombination of functional domains, particularly the receptor-binding domain, shapes the structural evolution of *Betacoronavirus* spike proteins. Although recombination is paramount in spike proteins, generating distinct evolutionary regimes across their domains, gradual mutational adaptation in the structure of the receptor-binding domain appears to play a dominant role in fine-tuning its specificity. The application of protein structure analysis provides a complementary perspective to phylogenetic studies of viral evolution, although it simultaneously exhibits advantages, redundancies, and limitations in capturing evolutionary dynamics compared to sequence-based approaches alone. These findings advance our understanding of the mechanisms underlying coronavirus emergence and highlight the importance of monitoring both sequence and structural evolution in viral surveillance efforts.

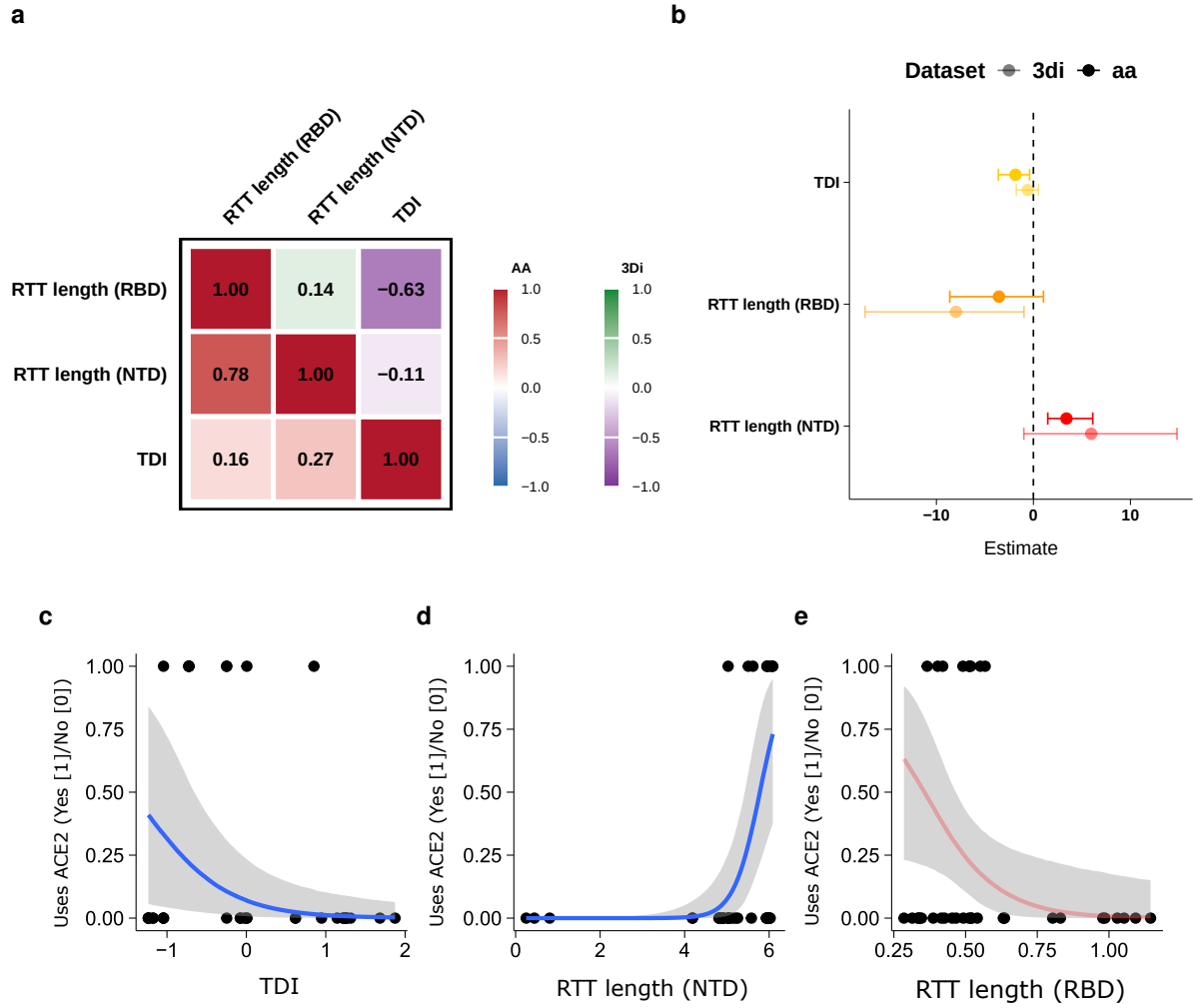


Figure 5. Analysing the influence of recombination vs. gradual mutation in the S1 subunit on ACE2 binding. **(a)** Pearson correlation matrix of the root-to-tip (RTT) length in molecular and structural trees covering the S1 NTD and RBD and their taxon discordance index (TDI; see [Methods](#)). **(b)** Bayesian regression estimates for predicting a taxon's propensity to bind ACE2. Errors bars show 95% confidence intervals (CIs). **(c, d, e)** Conditional effects plot of significant regressands: the taxon discordance index (TDI; **(c)**) and the NTD's root-to-tip length **(d)** in the AA dataset, and the RBD's root-to-tip length in the 3Di dataset **(e)**. Shaded areas represent 95% CIs.

Methods

Data

Data preparation. 2,113 betacoronavirus spike protein structures were retrieved from the Protein Data Bank (PDB) [56, 103] by augmenting the Cov3D database [55] (1,932 structures as of June 2025), a repository of experimentally obtained coronavirus structures. First, each existing spike protein structure in the database was manually verified by examining the metadata and the attached publications for missing data. Second, additional entries were incorporated by performing searches for similar sequences using MM-seqs2 [104] within the RCSB PDB API [103], using a sequence similarity cut-off of 50% for 6 structures: 6VXX (SARS-CoV-2, wildtype) [51], 9CRH (SARS-CoV-2, Delta) [105], 8WHS (SARS-CoV-2, Omicron BA.2.86) [106], 6NB6 (SARS-CoV) [65], 8IFN (MERS-CoV) [107], 9BSW (HCoV-HKU1) [108], 7SB3 (HCoV-OC43) [109], 9K75 (BANAL-103) [110]. This allowed us to

include 181 additional structures, including various understudied species within the betacoronavirus family [111]. All protein structures were downloaded in the mmCIF format using *biotite* [57]. For each viral strain in the metadata, its species and receptor were retrieved using NCBI taxonomy [112] and Google Scholar-powered (<https://scholar.google.com>) literature searches.

3Di sequence inference and cleaning. 3Di sequences were inferred in chain-wise fashion from 3D structures using Foldseek (v10.941cd33) [54]. 3Di sequences that did not pertain to spike protein chains (e.g., antibodies or receptors), did not encompass the entire spike protein, or were generally too short (< 850 residues) were discarded as likely to create artefacts during alignment. As spike glycoproteins are homotrimeric [21], heteromeric spike structures were also removed. The resulting dataset had 676 protein structures.

Sequence preprocessing. For the amino acid (AA) sequence set, alignment was performed using MAFFT (v7.5.26) [58, 113] with G-INS-I [114] and leavegappyout options. For the 3Di sequence set, alignment was performed using FAMSA-3Di [42], an adaptation of FAMSA (v2.2.3-99af4f9) [115] that uses the 3Di scoring matrix [54]. Similar to [42], alignments were trimmed using trimal (v1.5.rev0) [59], removing sites with 35% or more gaps. Both alignments were subsequently deduplicated, and sequences containing 50% or more gaps in the AA alignment were removed. The resulting alignments contained 257 sequences.

Finally, we used the following sampling procedure for downstream analyses. We selected the n sequences with the highest resolution protein structure for each virus, prioritising sequences not bound to antibodies or uncommon receptors to examine the structure of spike glycoprotein with minimal environmental influence [116, 117]. For SARS-CoV-2, one sequence was selected per major variant (Alpha, Delta, Omicron, Wildtype, and one sample representing other variants). From this, we generated two sub-datasets: one in which only a single sequence was retained per strain ($n = 1$, 36 sequences in total), and another in which up to three sequences were retained per strain ($n = 3$, 62 sequences in total). Despite their relatively small size compared to the raw data ($n = 2$, 113), these sequences are aligned, high-quality, and diverse, spanning three subgenera of betacoronaviruses (*Sarbecovirus*, *Merbecovirus*, and *Embecovirus*), and a variety of target receptors (see Table 1).

Phylogenetic modelling

For both AA and 3Di datasets, the 62-sequence subsets ('subset@3' in Fig. 1) were used to select an appropriate substitution model for all downstream analyses. Model selection was performed according to the Bayesian Information Criterion (BIC) using ModelFinder (MFP) [62] within the IQ-TREE toolkit (v3.0.1) [138]. For the AA alignment, model selection was performed on an augmented set from Puente-Lelievre *et al.* [42], adding empirical viral substitution matrices determined from *Orthomyxoviridae* [139] and *Flaviviridae* [140] proteins, as well as a general matrix estimated from the Pfam protein database (Q.pfam) [141]. The selected model was WAG+F+R4, denoting the empirical WAG model [142] with empirical base frequencies (F) and a FreeRate model [143, 144] with four categories (R4) (see Fig. S1). For the 3Di alignment, three substitution models were assessed: the 3Di substitution matrix used by Puente-Lelievre *et al.* [42], derived from the Foldseek 3Di scoring matrix [54], and two new general substitution matrices for 3Di sequences [47] determined using QMaker [141] from 3Di sequences inferred using the protein language model ProstT5 [145] (Q.3Di.LLM) or from Foldseek-clustered AlphaFold predictions (Q.3Di.AF). Drawing from previous model selection experiments with 3Di alignments [42], AA models were not fitted to the 3Di data, and vice versa. For the 3Di alignment, the best model according to ModelFinder was Q.3Di.AF+F+R4 (Table S2).

Thereafter, we cut the 36-sequence subsets ('subset@1' in Fig. 1) into m -chunks with a chunk size of 150 residues ($m = 8$ for AA, $m = 7$ for 3Di). Subsequently, we ran IQ-TREE to infer a single tree for each chunked alignment. The m trees were then used for mixture modelling across sites and trees using MAST [61] with the full multiple sequence alignment (MSA) as input. We then repeated the MAST analysis using the top $m - 1, \dots, 2$ trees with respect to their tree log-likelihoods in the initial run. This allowed us to select

the optimal number of classes m_{best} by comparing the BIC [146] of each MAST run, whereby we displayed the preferable tree mixtures and site likelihoods of the best model. For the latter, we displayed for each site the relative likelihood ΔL of the tree with the highest likelihood (Fig. 3b, 3e), calculated as:

$$\Delta L_{i,j} = L_{i,j} - \frac{1}{m_{best} - 1} \sum_{k \neq i} L_k$$

where $L_{i,j}$ is the log-likelihood of tree i at site j .

Post-processing

For each AA sequence in our subset, we inferred the position of the different domains using InterProScan (v5.75) [63] to overlay the class probabilities of each site on the different viral domains. The site coordinates obtained in the AA set were converted to 3Di coordinates using a custom script (see [Code availability](#)). For each tree in the optimal MAST run according to the BIC, we derived several tree statistics for each tree in the best model: tree length (the sum of all branch lengths), *stemminess* (the proportion of tree length attributed to internal branches) [147, 148], and *clock-likeness* (measured as the coefficient of variance of root-to-tip distances) [64, 78].

Assessing predictors of ACE2 usage

Using the trees best representing the N-terminal (NTD) and receptor-binding (RBD) domains in the AA and 3Di sets, we first visually examined topological discordance in the trees using co-phylogenetic plots using phytools (v2.4) [149] and ape (v5.8) [150] (Fig. 4a, 4b, S1a, S1). Subsequently, we assessed the relationship between each taxon's contribution to phylogenetic discordance between the NTD and RBD trees and its host receptor (ACE2 or not). To that end, we fitted a Bayesian logistic regression with brms (v2.22) [90] in R, specifying the following model:

$$\mathbb{I}_{j \text{ uses ACE2}} = \alpha + \sum_{i \in \{\text{RBD, NTD}\}} \beta_i \rho_{i,j} + \beta_{\text{TDI}} c_j$$

j denotes a taxon in 'Subset@1', $\rho_{i,j}$ its root-to-tip length, α the intercept, and β scalar parameters to be estimated. c_j denotes the taxon discordance index (TDI). Inspired from the taxon influence index [13], it measures the contribution of a taxon j to tree discordance c_j , calculated by measuring the mutual clustering information (MCI; computed using the TreeDist package (v2.9.2) [89]) when j is pruned from both phylogenetic trees. Higher values indicate higher discordance between the RBD and NTD trees after pruning the taxon j . Models were estimated using Markov Chain Monte Carlo (MCMC) sampling with the No-U-Turn Sampler [151] (NUTS). We ran four independent chains, each with 4,000 iterations, discarding the first 2,000 iterations as warmup, yielding 8,000 post-warmup samples for inference (2,000 samples per chain). To facilitate efficient sampling and minimise divergent transitions, a target acceptance rate of 0.99 was chosen, and a maximum tree depth of 15 was selected for building during the trajectory phase of the Hamiltonian Monte Carlo sampler. Model fit was assessed by examining the effective sample sizes (ESS), Gelman-Rubin statistic (Tables S3, S4), caterpillar plots (Fig. S2, S3), and posterior predictive checks (Fig. S4a, S5a).

Table 1. Distribution of *Betacoronavirus* strains and variants in the study. For SARS-CoV-2, “NS” indicates that the variant was or could not be determined with certainty. In Cov3D [55], they are typically named as “Orig” or “none”, but the exact meaning of these labels is uncertain. Consequently, these structures were ignored in downstream analyses. Three strains (BatCoV Vs-CoV 1, BtSy1, SE-PangolinCoV MjHKU4r-Cov-1) were not present in the final dataset as only having partial spike structures. Two strains (MERS-like BtCoV/Ii/GD/2014-422, SE-PangolinCoV MjHKU4r-Cov-1) were excluded due to sequence lengths exceeding the 99th percentile of the dataset, which would hinder alignment. *Sarb.*: *Sarbecovirus*; *Merb.*: *Merbecovirus*; *Emb.*: *Embecovirus*. ACE2: angiotensin-converting enzyme 2. DPP4: Dipeptidyl peptidase-4. 9-Oc-Ac-Sia: 9-O-acetylated sialic acid. CEACAM1: Carcinoembryonic antigen-related cell adhesion molecule 1.

Strain	Subgenus	Receptor	Count	Ref.	Strain	Subgenus	Receptor	Count	Ref.
SARS-CoV-2 (NS)	<i>Sarb.</i>	ACE2	1102	[75, 118, 119]	BatCoV CX1	<i>Sarb.</i>	ACE2	1	[120]
SARS-CoV-2 (Omicron)	<i>Sarb.</i>	ACE2	404	[75, 118, 119]	BatCoV HKU25	<i>Merb.</i>	ACE2	1	[121]
SARS-CoV-2 (Other)	<i>Sarb.</i>	ACE2	184	[75, 118, 119]	BatCoV HKU5-20	<i>Merb.</i>	ACE2	1	[122]
SARS-CoV-2 (Delta)	<i>Sarb.</i>	ACE2	100	[75, 118, 119]	BatCoV RmyN02	<i>Sarb.</i>	Unknown (ACE2 unlikely)	1	[123]
SARS-CoV	<i>Sarb.</i>	ACE2	75	[124]	BatCoV Vs-CoV 1	<i>Merb.</i>	Unknown (not DPP4)	1	[125]
MERS-CoV	<i>Merb.</i>	DPP4	70	[126]	BtCoV-422	<i>Merb.</i>	DPP4	1	[125]
HCoV-HKU1	<i>Emb.</i>	9-O-Ac-Sia	48	[127]	BtRF-BetaCoV/HeB2013	<i>Sarb.</i>	Unknown (ACE2 unlikely)	1	[128]
SARS-CoV-2 (Alpha)	<i>Sarb.</i>	ACE2	43	[75, 118, 119]	BtRs-BetaCoV/BtKY72	<i>Sarb.</i>	ACE2	1	[129]
SARS-CoV-2 (Wildtype)	<i>Sarb.</i>	ACE2	22	[75, 118, 119]	BtRs-BetaCoV/GX2013	<i>Sarb.</i>	Unknown (ACE2 unlikely)	1	[128]
HCoV-OC43	<i>Emb.</i>	9-O-Ac-Sia	14	[127]	BtRs-BetaCoV/YN2013	<i>Sarb.</i>	Unknown (ACE2 unlikely)	1	[130]
Bat SL-CoV-WIV1	<i>Sarb.</i>	ACE2	6	[118]	BtSY1	<i>Sarb.</i>	Unknown (ACE2 likely)	1	[131]
BatCoV HKU5	<i>Merb.</i>	ACE2	4	[122]	Civet SARS CoV 007	<i>Sarb.</i>	ACE2	1	[132]
BatCoV BANAL-20-236	<i>Sarb.</i>	ACE2	3	[92]	Civet SARS CoV SZ3	<i>Sarb.</i>	ACE2	1	[132]
BatCoV BANAL-20-52	<i>Sarb.</i>	ACE2	3	[92]	EU-HedgehogCoV	<i>Merb.</i>	Unknown (not DPP4)	1	[125]
BatCoV RaTG13	<i>Sarb.</i>	ACE2	3	[88, 133]	MERS-like BtCoV/Ii/GD/2014-422	<i>Merb.</i>	DPP4	1	[125]
BatCoV SHC014	<i>Sarb.</i>	ACE2	3	[118]	MERS-like PDF-2180	<i>Merb.</i>	ACE2	1	[134]
CN-HedgehogCoV HKU31	<i>Merb.</i>	Unknown (not DPP4)	2	[125]	PCoV-GD	<i>Sarb.</i>	ACE2	1	[135]
MCoV A59	<i>Emb.</i>	CEACAM1	2	[136]	PCoV-GX	<i>Sarb.</i>	ACE2	1	[135]
BatCoV BANAL-103	<i>Sarb.</i>	ACE2	1	[92]	PRD-0038	<i>Sarb.</i>	ACE2	1	[137]
BatCoV BANAL-52	<i>Sarb.</i>	ACE2	1	[92]	SA-BatCoV	<i>Merb.</i>	Unknown (not DPP4)	1	[125]
BatCoV BM48-31	<i>Sarb.</i>	Unknown (not ACE2)	1	[25, 129]	SE-PangolinCoV MjHKU4r-Cov-1	<i>Merb.</i>	DPP4	1	[125]

Implementation

All steps in the pipeline were written with custom scripts in Python (v3.13.7) and R (v4.5) and version-controlled for reproducibility using Data Version Control (DVC) (<https://dvc.org>). Code to reproduce all steps in the pipeline shown in Fig. 1 is available at <https://github.com/Neclow/bcov3d>.

Data availability All data and trees are available online in Zenodo [152].

Code availability All code relevant to reproduce the experiments is available online at <https://github.com/Neclow/bcov3d>.

References

- Rota, P. A. *et al.* Characterization of a novel coronavirus associated with severe acute respiratory syndrome. *Science* **300**, 1394–1399 (2003).
- Drosten, C. *et al.* Identification of a novel coronavirus in patients with severe acute respiratory syndrome. *New England journal of medicine* **348**, 1967–1976 (2003).
- Ksiazek, T. G. *et al.* A novel coronavirus associated with severe acute respiratory syndrome. *New England journal of medicine* **348**, 1953–1966 (2003).
- Zaki, A. M., Van Boheemen, S., Bestebroer, T. M., Osterhaus, A. D. & Fouchier, R. A. Isolation of a novel coronavirus from a man with pneumonia in Saudi Arabia. *New England journal of medicine* **367**, 1814–1820 (2012).
- Huang, C. *et al.* Clinical features of patients infected with 2019 novel coronavirus in Wuhan, China. *The lancet* **395**, 497–506 (2020).
- Zhou, P. *et al.* A pneumonia outbreak associated with a new coronavirus of probable bat origin. *Nature* **579**, 270–273 (2020).
- Zhu, N. *et al.* A novel coronavirus from patients with pneumonia in China, 2019. *New England journal of medicine* **382**, 727–733 (2020).
- Paules, C. I., Marston, H. D. & Fauci, A. S. Coronavirus infections—more than just the common cold. *JAMA : the journal of the American Medical Association* **323**, 707–708 (2020).
- Liu, D. X., Liang, J. Q. & Fung, T. S. Human coronavirus-229E, -OC43, -NL63, and -HKU1 (coronaviridae). *Encyclopedia of virology*, 428 (2021).
- Li, W. *et al.* Bats Are Natural Reservoirs of SARS-Like Coronaviruses. *Science* **310**, 676–679. doi:10.1126/science.1118391 (2005).
- Ruiz-Aravena, M. *et al.* Ecology, evolution and spillover of coronaviruses from bats. *Nature Reviews Microbiology* **20**, 299–314. doi:10.1038/s41579-021-00652-2 (2022).
- Wang, W. *et al.* Extensive genetic diversity and host range of rodent-borne coronaviruses. *Virus Evolution* **6**, veaa078. doi:10.1093/ve/veaa078 (2020).
- Marani, M., Katul, G. G., Pan, W. K. & Parolari, A. J. Intensity and frequency of extreme novel epidemics. doi:10.1073/pnas.2105482118 (2021).
- Vora, N. M. *et al.* Want to prevent pandemics? Stop spillovers. *Nature* **605**, 419–422. doi:10.1038/d41586-022-01312-y (2022).

15. Ge, X.-Y. *et al.* Isolation and characterization of a bat SARS-like coronavirus that uses the ACE2 receptor. *Nature* **503**, 535–538. doi:[10.1038/nature12711](https://doi.org/10.1038/nature12711) (2013).
16. Azhar, E. I. *et al.* Evidence for camel-to-human transmission of MERS coronavirus. *New England Journal of Medicine* **370**, 2499–2505. doi:[10.1056/NEJMoa1401505](https://doi.org/10.1056/NEJMoa1401505) (2014).
17. Cui, J., Li, F. & Shi, Z.-L. Origin and evolution of pathogenic coronaviruses. *Nature Reviews Microbiology* **17**, 181–192. doi:[10.1038/s41579-018-0118-9](https://doi.org/10.1038/s41579-018-0118-9) (2019).
18. Lam, T. T.-Y. *et al.* Identifying SARS-CoV-2-related coronaviruses in Malayan pangolins. *Nature* **583**, 282–285. doi:[10.1038/s41586-020-2169-0](https://doi.org/10.1038/s41586-020-2169-0) (2020).
19. Tan, C. C. S. *et al.* Transmission of SARS-CoV-2 from humans to animals and potential host adaptation. *Nature Communications* **13**, 2988. doi:[10.1038/s41467-022-30698-6](https://doi.org/10.1038/s41467-022-30698-6) (2022).
20. Jackson, C. B., Farzan, M., Chen, B. & Choe, H. Mechanisms of SARS-CoV-2 entry into cells. *Nature reviews Molecular cell biology* **23**, 3–20 (2022).
21. Li, F. Structure, function, and evolution of coronavirus spike proteins. *Annual review of virology* **3**, 237–261 (2016).
22. Belouzard, S., Millet, J. K., Licitra, B. N. & Whitaker, G. R. Mechanisms of coronavirus cell entry mediated by the viral spike protein. *Viruses* **4**, 1011–1033 (2012).
23. Li, F. Receptor recognition mechanisms of coronaviruses: a decade of structural studies. *Journal of virology* **89**, 1954–1964 (2015).
24. Letko, M. *et al.* Adaptive Evolution of MERS-CoV to Species Variation in DPP4. *Cell Reports* **24**, 1730–1737. doi:[10.1016/j.celrep.2018.07.045](https://doi.org/10.1016/j.celrep.2018.07.045) (2018).
25. Starr, T. N. *et al.* ACE2 binding is an ancestral and evolvable trait of sarbecoviruses. *Nature* **603**, 913–918. doi:[10.1038/s41586-022-04464-z](https://doi.org/10.1038/s41586-022-04464-z) (2022).
26. Lytras, S. *et al.* Exploring the Natural Origins of SARS-CoV-2 in the Light of Recombination. *Genome Biology and Evolution* **14**, evac018. doi:[10.1093/gbe/evac018](https://doi.org/10.1093/gbe/evac018) (2022).
27. Wells, H. L. *et al.* The coronavirus recombination pathway. *Cell Host & Microbe* **31**, 874–889. doi:[10.1016/j.chom.2023.05.003](https://doi.org/10.1016/j.chom.2023.05.003) (2023).
28. Su, S. *et al.* Epidemiology, genetic recombination, and pathogenesis of coronaviruses. *Trends in microbiology* **24**, 490–502 (2016).
29. Forni, D., Cagliani, R. & Sironi, M. Recombination and positive selection differentially shaped the diversity of betacoronavirus subgenera. *Viruses* **12**, 1313 (2020).
30. De Klerk, A. *et al.* Conserved recombination patterns across coronavirus subgenera. *Virus Evolution* **8**, veac054. doi:[10.1093/ve/veac054](https://doi.org/10.1093/ve/veac054) (2022).
31. Bobay, L.-M., O'Donnell, A. C. & Ochman, H. Recombination events are concentrated in the spike protein region of Betacoronaviruses. *PLoS Genetics* **16**, e1009272 (2020).
32. Turakhia, Y. *et al.* Pandemic-scale phylogenomics reveals the SARS-CoV-2 recombination landscape. *Nature* **609**, 994–997 (2022).
33. Patiño-Galindo, J. Á. *et al.* Recombination and lineage-specific mutations linked to the emergence of SARS-CoV-2. doi:[10.1101/2020.02.10.942748](https://doi.org/10.1101/2020.02.10.942748) (2021).
34. Mannar, D. *et al.* Altered receptor binding, antibody evasion and retention of T cell recognition by the SARS-CoV-2 XBB.1.5 spike protein. *Nature Communications* **15**, 1854. doi:[10.1038/s41467-024-46104-2](https://doi.org/10.1038/s41467-024-46104-2) (2024).
35. Vöhringer, H. S. *et al.* Genomic reconstruction of the SARS-CoV-2 epidemic in England. *Nature* **600**, 506–511. doi:[10.1038/s41586-021-04069-y](https://doi.org/10.1038/s41586-021-04069-y) (2021).
36. Obermeyer, F. *et al.* Analysis of 6.4 million SARS-CoV-2 genomes identifies mutations associated with fitness. *Science* **376**, 1327–1332 (2022).
37. Khurana, M. P. *et al.* High-resolution epidemiological landscape from 290,000 SARS-CoV-2 genomes from Denmark. *Nature Communications* **15**, 7123 (2024).
38. Jumper, J. *et al.* Highly accurate protein structure prediction with AlphaFold. *Nature* **596**, 583–589 (2021).
39. Lin, Z. *et al.* Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science* **379**, 1123–1130 (2023).
40. Abramson, J. *et al.* Accurate structure prediction of biomolecular interactions with AlphaFold 3. *Nature* **630**, 493–500 (2024).
41. Moi, D. *et al.* Structural phylogenetics unravels the evolutionary diversification of communication systems in gram-positive bacteria and their viruses. *bioRxiv : the preprint server for biology*, 2023–09 (2023).
42. Puente-Lelievre, C. *et al.* Tertiary-interaction characters enable fast, model-based structural phylogenetics beyond the twilight zone. doi:[10.1101/2023.12.12.571181](https://doi.org/10.1101/2023.12.12.571181) (2024).
43. Fullmer, M. S., Puente-Lelievre, C. & Matzke, N. J. Adding 3Di characters to amino acid datasets can improve resolution, but the effect is weaker in shorter and alpha-helical proteins such as histones. *bioRxiv : the preprint server for biology*, 2025–06 (2025).
44. Mifsud, J. C. O. *et al.* Mapping glycoprotein structure reveals Flaviviridae evolutionary history. *Nature* **633**, 695–703. doi:[10.1038/s41586-024-07899-8](https://doi.org/10.1038/s41586-024-07899-8) (2024).
45. Mifsud, J. C. O., Suchard, M. A., Holmes, E. C. & Lemey, P. Recent advances in the inference of deep viral evolutionary history. *Journal of Virology* **0**, e00292–25. doi:[10.1128/jvi.00292-25](https://doi.org/10.1128/jvi.00292-25) (2025).
46. Puente-Lelievre, C., Malik, A. & Douglas, J. Protein Structural Phylogenetics. *Genome Biology and Evolution* **17**, evaf139. doi:[10.1093/gbe/evaf139](https://doi.org/10.1093/gbe/evaf139) (2025).

47. Garg, S. G. & Hochberg, G. K. A general substitution matrix for structural phylogenetics. *Molecular Biology and Evolution* **42**, msaf124 (2025).
48. Mutti, G., Ocaña-Pallarès, E. & Gabaldón, T. Newly developed structure-based methods do not outperform standard sequence-based methods for large-scale phylogenomics. *Molecular Biology and Evolution*, msaf149 (2025).
49. Crick, F. H. *On protein synthesis in Symp soc exp biol* **12**. Number: 138-63 (1958), 8.
50. Li, F., Li, W., Farzan, M. & Harrison, S. C. Structure of SARS Coronavirus Spike Receptor-Binding Domain Complexed with Receptor. *Science*. doi:[10.1126/science.1116480](https://doi.org/10.1126/science.1116480) (2005).
51. Walls, A. C. *et al.* Structure, function, and antigenicity of the SARS-CoV-2 spike glycoprotein. *Cell* **181**, 281–292 (2020).
52. Lan, J. *et al.* Structural insights into the SARS-CoV-2 Omicron RBD-ACE2 interaction. *Cell Research* **32**, 593–595. doi:[10.1038/s41422-022-00644-8](https://doi.org/10.1038/s41422-022-00644-8) (2022).
53. Jiaming, L. *et al.* The recombinant N-terminal domain of spike proteins is a potential vaccine against Middle East respiratory syndrome coronavirus (MERS-CoV) infection. *Vaccine* **35**, 10–18. doi:[10.1016/j.vaccine.2016.11.064](https://doi.org/10.1016/j.vaccine.2016.11.064) (2017).
54. Van Kempen, M. *et al.* Fast and accurate protein structure search with Foldseek. *Nature biotechnology* **42**, 243–246 (2024).
55. Gowthaman, R. *et al.* CoV3D: a database of high resolution coronavirus protein structures. *Nucleic acids research* **49**, D282–D287 (2021).
56. Rose, P. W. *et al.* The RCSB Protein Data Bank: redesigned web site and web services. *Nucleic acids research* **39**, D392–D401 (2010).
57. Kunzmann, P. & Hamacher, K. Biotite: a unifying open source computational biology framework in Python. *BMC bioinformatics* **19**, 346 (2018).
58. Katoh, K. & Standley, D. M. MAFFT multiple sequence alignment software version 7: improvements in performance and usability. *Molecular biology and evolution* **30**, 772–780 (2013).
59. Capella-Gutiérrez, S., Silla-Martínez, J. M. & Gabaldón, T. trimAl: a tool for automated alignment trimming in large-scale phylogenetic analyses. *Bioinformatics (Oxford, England)* **25**, 1972–1973 (2009).
60. Minh, B. Q. *et al.* IQ-TREE 2: new models and efficient methods for phylogenetic inference in the genomic era. *Molecular biology and evolution* **37**, 1530–1534 (2020).
61. Wong, T. K. *et al.* MAST: phylogenetic inference with mixtures across sites and trees. *Systematic biology* **73**, 375–391 (2024).
62. Kalyanamoorthy, S., Minh, B. Q., Wong, T. K., Von Haeseler, A. & Jermiin, L. S. ModelFinder: fast model selection for accurate phylogenetic estimates. *Nature methods* **14**, 587–589 (2017).
63. Jones, P. *et al.* InterProScan 5: genome-scale protein function classification. *Bioinformatics (Oxford, England)* **30**, 1236–1240 (2014).
64. Zuckerkandl, E. & Pauling, L. B. *Molecular disease, evolution, and genic heterogeneity in Horizons in biochemistry: Albert szent-györgyi dedicatory volume* (eds Kasha, M. & Pullman, B.) 189–225 (Academic Press, New York, 1962).
65. Walls, A. C. *et al.* Unexpected receptor functional mimicry elucidates activation of coronavirus fusion. *Cell* **176**, 1026–1039 (2019).
66. Sehna, D. *et al.* Mol* Viewer: modern web app for 3D visualization and analysis of large biomolecular structures. *Nucleic acids research* **49**, W431–W437 (2021).
67. Boussau, B., Guéguen, L. & Gouy, M. A Mixture Model and a Hidden Markov Model to Simultaneously Detect Recombination Breakpoints and Reconstruct Phylogenies. *Evolutionary Bioinformatics Online* **5**, 67–79. doi:[10.4137/ebo.s2242](https://doi.org/10.4137/ebo.s2242) (2009).
68. Kim, R. S., Levy Karin, E., Mirdita, M., Chikhi, R. & Steinegger, M. BFVD—a large repository of predicted viral protein structures. *Nucleic acids research* **53**, D340–D347 (2025).
69. Litvin, U. *et al.* Viro3D: a comprehensive database of virus protein structure predictions. *bioRxiv : the preprint server for biology*, 2024–12 (2024).
70. Magana Gomez, P. G. & Kovalevskiy, O. *AlphaFold: A practical guide* 2024. doi:[10.6019/tol.alphafold-w.2024.00001.1](https://doi.org/10.6019/tol.alphafold-w.2024.00001.1).
71. Lin, P.-Y. *et al.* Analysing protein complexes in plant science: insights and limitation with AlphaFold 3. *Botanical Studies* **66**, 14 (2025).
72. Wayment-Steele, H. K. *et al.* Predicting multiple conformations via sequence clustering and AlphaFold2. *Nature* **625**, 832–839 (2024).
73. Wee, J. & Wei, G.-W. Rapid response to fast viral evolution using AlphaFold 3-assisted topological deep learning. *Virus Evolution* **11**, veaf026 (2025).
74. Woo, P. C. Y. *et al.* ICTV Virus Taxonomy Profile: Coronaviridae 2023. *Journal of General Virology* **104**, 001843. doi:[10.1099/jgv.0.001843](https://doi.org/10.1099/jgv.0.001843) (2023).
75. Lan, J. *et al.* Structure of the SARS-CoV-2 spike receptor-binding domain bound to the ACE2 receptor. *Nature* **581**, 215–220 (2020).
76. Huang, Y., Yang, C., Xu, X.-f., Xu, W. & Liu, S.-w. Structural and functional properties of SARS-CoV-2 spike protein: potential antiviral drug development for COVID-19. *Acta Pharmacologica Sinica* **41**, 1141–1149 (2020).
77. Illergård, K., Ardell, D. H. & Elofsson, A. Structure is three to ten times more conserved than sequence—a study of structural response in protein cores. *Proteins: Structure, Function, and Bioinformatics* **77**, 499–508 (2009).

78. Zuckerkandl, E. & Pauling, L. *Evolutionary divergence and convergence in proteins in Evolving genes and proteins* 97–166 (Elsevier, 1965).
79. Kubo, H., Yamada, Y. K. & Taguchi, F. Localization of neutralizing epitopes and the receptor-binding site within the amino-terminal 330 amino acids of the murine coronavirus spike protein. *Journal of Virology* **68**, 5403–5410. doi:[10.1128/jvi.68.9.5403-5410.1994](https://doi.org/10.1128/jvi.68.9.5403-5410.1994) (1994).
80. Peng, G. *et al.* Crystal structure of mouse coronavirus receptor-binding domain complexed with its murine receptor. *Proceedings of the National Academy of Sciences* **108**, 10696–10701. doi:[10.1073/pnas.1104306108](https://doi.org/10.1073/pnas.1104306108) (2011).
81. Shang, J. *et al.* Structure of mouse coronavirus spike protein complexed with receptor reveals mechanism for viral entry. *PLOS Pathogens* **16**, 1–20. doi:[10.1371/journal.ppat.1008392](https://doi.org/10.1371/journal.ppat.1008392) (2020).
82. Boni, M. F. *et al.* Evolutionary origins of the SARS-CoV-2 sarbecovirus lineage responsible for the COVID-19 pandemic. *Nature microbiology* **5**, 1408–1417 (2020).
83. Guo, H. *et al.* The Glycan-Binding Trait of the Sarbecovirus Spike N-Terminal Domain Reveals an Evolutionary Footprint. *Journal of Virology* **96**, e00958–22. doi:[10.1128/jvi.00958-22](https://doi.org/10.1128/jvi.00958-22) (2022).
84. Tang, X., Qian, Z., Lu, X. & Lu, J. Adaptive evolution of the spike protein in coronaviruses. *Molecular Biology and Evolution* **40**, msad089 (2023).
85. Tolentino, J. E., Lytras, S., Ito, J. & Sato, K. Recombination analysis on the receptor switching event of MERS-CoV and its close relatives: implications for the emergence of MERS-CoV. *Virology Journal* **21**, 84 (2024).
86. Bosch, B. J., van der Zee, R., de Haan, C. A. M. & Rottier, P. J. M. The Coronavirus Spike Protein Is a Class I Virus Fusion Protein: Structural and Functional Characterization of the Fusion Core Complex. *Journal of Virology* **77**, 8801–8811. doi:[10.1128/jvi.77.16.8801-8811.2003](https://doi.org/10.1128/jvi.77.16.8801-8811.2003) (2003).
87. De Groot, R. J. *et al.* Evidence for a coiled-coil structure in the spike proteins of coronaviruses. *Journal of Molecular Biology* **196**, 963–966. doi:[10.1016/0022-2836\(87\)90422-0](https://doi.org/10.1016/0022-2836(87)90422-0) (1987).
88. Wrobel, A. G. *et al.* SARS-CoV-2 and bat RaTG13 spike glycoprotein structures inform on virus evolution and furin-cleavage effects. *Nature structural & molecular biology* **27**, 763–767 (2020).
89. Smith, M. R. Information theoretic Generalized Robinson-Foulds metrics for comparing phylogenetic trees. *Bioinformatics (Oxford, England)* **36**, 5007–5013. doi:[10.1093/bioinformatics/btaa614](https://doi.org/10.1093/bioinformatics/btaa614) (2020).
90. Bürkner, P.-C. brms: An R package for Bayesian multilevel models using Stan. *Journal of Statistical Software* **80**, 1–28 (2017).
91. Li, X. *et al.* Emergence of SARS-CoV-2 through recombination and strong purifying selection. *Science advances* **6**, eabb9153 (2020).
92. Temmam, S. *et al.* Bat coronaviruses related to SARS-CoV-2 and infectious for human cells. *Nature* **604**, 330–336 (2022).
93. Qing, E. *et al.* Dynamics of SARS-CoV-2 Spike Proteins in Cell Entry: Control Elements in the Amino-Terminal Domains. *mBio*. doi:[10.1128/mbio.01590-21](https://doi.org/10.1128/mbio.01590-21) (2021).
94. Meng, B. *et al.* SARS-CoV-2 spike N-terminal domain modulates TMPRSS2-dependent viral entry and fusogenicity. *Cell Reports* **40**, 111220. doi:[10.1016/j.celrep.2022.111220](https://doi.org/10.1016/j.celrep.2022.111220) (2022).
95. Wells, H. L. *et al.* The evolutionary history of ACE2 usage within the coronavirus subgenus Sarbecovirus. *Virus Evolution* **7**, veab007. doi:[10.1093/ve/veab007](https://doi.org/10.1093/ve/veab007) (2021).
96. Wrobel, A. G. *et al.* Structure and binding properties of Pangolin-CoV spike glycoprotein inform the evolution of SARS-CoV-2. *Nature Communications* **12**, 837. doi:[10.1038/s41467-021-21006-9](https://doi.org/10.1038/s41467-021-21006-9) (2021).
97. Aiweesakun, P. & Katzourakis, A. Time-Dependent Rate Phenomenon in Viruses. *Journal of Virology* **90**, 7184–7195. doi:[10.1128/jvi.00593-16](https://doi.org/10.1128/jvi.00593-16) (2016).
98. Markov, P. V. *et al.* The evolution of SARS-CoV-2. *Nature Reviews Microbiology* **21**, 361–379 (2023).
99. Zheng, Y. *et al.* SARS-CoV-2 NSP5 and N protein counteract the RIG-I signaling pathway by suppressing the formation of stress granules. *Signal Transduction and Targeted Therapy* **7**, 22. doi:[10.1038/s41392-022-00878-3](https://doi.org/10.1038/s41392-022-00878-3) (2022).
100. Borgogna, C. *et al.* Convergent evolution of the G3BP1-binding motif in betacoronavirus nucleocapsid proteins. *Virus Evolution*, veaf063. doi:[10.1093/ve/veaf063](https://doi.org/10.1093/ve/veaf063) (2025).
101. Minkoff, J. M. & tenOever, B. Innate immune evasion strategies of SARS-CoV-2. *Nature Reviews Microbiology* **21**, 178–194. doi:[10.1038/s41579-022-00839-1](https://doi.org/10.1038/s41579-022-00839-1) (2023).
102. Dutta, N. K., Mazumdar, K. & Gordy, J. T. The Nucleocapsid Protein of SARS-CoV-2: a Target for Vaccine Development. *Journal of Virology* **94**, 10.1128/jvi.00647–20. doi:[10.1128/jvi.00647-20](https://doi.org/10.1128/jvi.00647-20) (2020).
103. Berman, H. M. *et al.* The protein data bank. *Nucleic acids research* **28**, 235–242 (2000).
104. Steinegger, M. & Söding, J. MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets. *Nature Biotechnology* **35**, 1026–1028. doi:[10.1038/nbt.3988](https://doi.org/10.1038/nbt.3988) (2017).
105. Ke, Z. *et al.* Virion morphology and on-virus spike protein structures of diverse SARS-CoV-2 variants. *The EMBO Journal* **43**, 6469–6495 (2024).
106. Liu, P. *et al.* Spike N354 glycosylation augments SARS-CoV-2 fitness for human adaptation through

- structural plasticity. *National Science Review* **11**, nwae206 (2024).
107. Tian, L., Wang, R., Zhang, L. & Wang, X. Structures and neutralizing mechanisms of camel nanobodies targeting the receptor-binding domain of MERS-CoV spike glycoprotein 2024. doi:[10.2210/pdb8IFN/pdb](https://doi.org/10.2210/pdb8IFN/pdb).
 108. Jin, M. *et al.* Human coronavirus HKU1 spike structures reveal the basis for sialoglycan specificity and carbohydrate-promoted conformational changes. *Nature Communications* **16**, 4158 (2025).
 109. Bangaru, S. *et al.* Structural mapping of antibody landscapes to human betacoronavirus spike proteins. *Science Advances* **8**, eabn2911 (2022).
 110. Li, Q. *et al.* Structural and functional constraints on spike activation and host protease utilization limit cell entry of SARS-CoV-2-related bat coronaviruses. *Journal of Virology*. doi:[10.1128/jvi.01007-25](https://doi.org/10.1128/jvi.01007-25) (2025).
 111. Hu, B., Guo, H., Zhou, P. & Shi, Z.-L. Characteristics of SARS-CoV-2 and COVID-19. *Nature reviews microbiology* **19**, 141–154 (2021).
 112. Schoch, C. L. *et al.* NCBI Taxonomy: a comprehensive update on curation, resources and tools. *Database-the Magazine of Electronic Database Reviews* **2020**, baaa062 (2020).
 113. Katoh, K. & Standley, D. M. A simple method to control over-alignment in the MAFFT multiple sequence alignment program. *Bioinformatics* **32**, 1933–1942. doi:[10.1093/bioinformatics/btw108](https://doi.org/10.1093/bioinformatics/btw108) (2016).
 114. Needleman, S. B. & Wunsch, C. D. A general method applicable to the search for similarities in the amino acid sequence of two proteins. *Journal of Molecular Biology* **48**, 443–453. doi:[10.1016/0022-2836\(70\)90057-4](https://doi.org/10.1016/0022-2836(70)90057-4) (1970).
 115. Deorowicz, S., Debudaj-Grabysz, A. & Gudyś, A. FAMSA: Fast and accurate multiple sequence alignment of huge protein families. *Scientific reports* **6**, 33964 (2016).
 116. Pronker, M. F. *et al.* Sialoglycan binding triggers spike opening in a human coronavirus. *Nature* **624**, 201–206. doi:[10.1038/s41586-023-06599-z](https://doi.org/10.1038/s41586-023-06599-z) (2023).
 117. Hasse, T. *et al.* Mechanistic insights into ligand dissociation from the SARS-CoV-2 spike glycoprotein. *PLOS Computational Biology* **20**, e1011955. doi:[10.1371/journal.pcbi.1011955](https://doi.org/10.1371/journal.pcbi.1011955) (2024).
 118. Letko, M., Marzi, A. & Munster, V. Functional assessment of cell entry and receptor usage for SARS-CoV-2 and other lineage B betacoronaviruses. *Nature microbiology* **5**, 562–569 (2020).
 119. Shang, J. *et al.* Structural basis of receptor recognition by SARS-CoV-2. *Nature* **581**, 221–224 (2020).
 120. Xu, Z. *et al.* CX1/BtSY2 and BANAL-20-52 exhibit broader receptor binding and higher affinities to multiple animal ACE2 orthologs than SARS-CoV-2 prototype. *Journal of Virology*, e00283–25 (2025).
 121. Liu, C. *et al.* ACE2 utilization of HKU25 clade MERS-related coronaviruses with broad geographic distribution. *bioRxiv : the preprint server for biology*. doi:[10.1101/2025.02.19.639017](https://doi.org/10.1101/2025.02.19.639017) (2025).
 122. Catanzaro, N. J. *et al.* ACE2 from *Pipistrellus abramus* bats is a receptor for HKU5 coronaviruses. *Nature Communications* **16**, 4932 (2025).
 123. Zhou, H. *et al.* A novel bat coronavirus closely related to SARS-CoV-2 contains natural insertions at the S1/S2 cleavage site of the spike protein. *Current biology* **30**, 2196–2203 (2020).
 124. Li, W. *et al.* Angiotensin-converting enzyme 2 is a functional receptor for the SARS coronavirus. *Nature* **426**, 450–454 (2003).
 125. Yuan, H. *et al.* Structures and receptor binding activities of merbecovirus spike proteins reveal key signatures for human DPP4 adaptation. *Science Advances* **11**, eadv7296 (2025).
 126. Raj, V. S. *et al.* Dipeptidyl peptidase 4 is a functional receptor for the emerging human coronavirus-EMC. *Nature* **495**, 251–254 (2013).
 127. Hulswit, R. J. *et al.* Human coronaviruses OC43 and HKU1 bind to 9-O-acetylated sialic acids via a conserved receptor-binding site in spike protein domain A. *Proceedings of the National Academy of Sciences* **116**, 2681–2690 (2019).
 128. Wang, J. *et al.* SARS-related coronavirus S-protein structures reveal synergistic RBM interactions underpinning high-affinity human ACE2 binding. *Science Advances* **11**, eadr8772 (2025).
 129. Su, C. *et al.* Structural characteristics of BtKY72 RBD bound to bat ACE2 reveal multiple key residues affecting ACE2 usage of sarbecoviruses. *MBio* **15**, e01404–24 (2024).
 130. Roelle, S. M., Shukla, N., Pham, A. T., Bruchez, A. M. & Matreyek, K. A. Expanded ACE2 dependencies of diverse SARS-like coronavirus receptor binding domains. *PLoS Biology* **20**, e3001738 (2022).
 131. Xu, Z. & Sun, J. *the structure of BtSY1_RBD/hACE2 protein* 2025. doi:[10.2210/pdb8WSR/pdb](https://doi.org/10.2210/pdb8WSR/pdb).
 132. Li, W. *et al.* Animal origins of the severe acute respiratory syndrome coronavirus: insight from ACE2-S-protein interactions. *Journal of virology* **80**, 4211–4219 (2006).
 133. Zhang, W. *et al.* Structural basis for mouse receptor recognition by bat SARS2-like coronaviruses. *Proceedings of the National Academy of Sciences* **121**, e2322600121 (2024).
 134. Xiong, Q. *et al.* Close relatives of MERS-CoV in bats use ACE2 as their functional receptors. *Nature* **612**, 748–757 (2022).
 135. Nie, J. *et al.* Functional comparison of SARS-CoV-2 with closely related pangolin and bat coronaviruses. *Cell discovery* **7**, 21 (2021).
 136. Ryu, S. *et al.* Ketogenic diet restrains aging-induced exacerbation of coronavirus infection in mice. *eLife* **10**, e66522. doi:[10.7554/eLife.66522](https://doi.org/10.7554/eLife.66522) (2021).

137. Lee, J. *et al.* Broad receptor tropism and immunogenicity of a clade 3 sarbecovirus. *Cell host & microbe* **31**, 1961–1973 (2023).
138. Wong, T. K. F. *et al.* IQ-TREE 3: Phylogenomic Inference Software using Complex Evolutionary Models (2025).
139. Dang, C. C., Le, Q. S., Gascuel, O. & Le, V. S. FLU, an amino acid substitution model for influenza proteins. *BMC Evolutionary Biology* **10**, 99. doi:[10.1186/1471-2148-10-99](https://doi.org/10.1186/1471-2148-10-99) (2010).
140. Le, T. K. & Vinh, L. S. FLAVI: An Amino Acid Substitution Model for Flaviviruses. *Journal of Molecular Evolution* **88**, 445–452. doi:[10.1007/s00239-020-09943-3](https://doi.org/10.1007/s00239-020-09943-3) (2020).
141. Minh, B. Q., Dang, C. C., Vinh, L. S. & Lanfear, R. QMaker: fast and accurate method to estimate empirical models of protein evolution. *Systematic Biology* **70**, 1046–1060 (2021).
142. Whelan, S. & Goldman, N. A general empirical model of protein evolution derived from multiple protein families using a maximum-likelihood approach. *Molecular biology and evolution* **18**, 691–699 (2001).
143. Yang, Z. A space-time process model for the evolution of DNA sequences. *Genetics* **139**, 993–1005 (1995).
144. Soubrier, J. *et al.* The influence of rate heterogeneity among sites on the time dependence of molecular rates. *Molecular biology and evolution* **29**, 3345–3358 (2012).
145. Heinzinger, M. *et al.* Bilingual language model for protein sequence and structure. *NAR Genomics and Bioinformatics* **6**, lqae150 (2024).
146. Schwarz, G. Estimating the dimension of a model. *The annals of statistics*, 461–464 (1978).
147. Fiala, K. L. & Sokal, R. R. Factors determining the accuracy of cladogram estimation: evaluation using computer simulation. *Evolution; international journal of organic evolution* **39**, 609–622 (1985).
148. Rohlf, F. J., Chang, W., Sokal, R. & Kim, J. Accuracy of estimated phylogenies: effects of tree topology and evolutionary model. *Evolution; international journal of organic evolution* **44**, 1671–1684 (1990).
149. Revell, L. J. hytools 2.0: an updated R ecosystem for phylogenetic comparative methods (and other things). *PeerJ* **12**, e16505. doi:[10.7717/peerj.16505](https://doi.org/10.7717/peerj.16505) (2024).
150. Paradis, E. & Schliep, K. ape 5.0: an environment for modern phylogenetics and evolutionary analyses in R. *Bioinformatics (Oxford, England)* **35**, 526–528 (2019).
151. Hoffman, M. D. & Gelman, A. The no-u-turn sampler: Adaptively setting path lengths in hamiltonian monte carlo. *Journal of Machine Learning Research* **15**, 1593–1623 (2014).
152. Scheidwasser, N. A. *Recombination of binding domains shapes structural evolution of Betacoronavirus spike proteins* 2025. doi:[10.5281/zenodo.17242359](https://doi.org/10.5281/zenodo.17242359).

Funding SB and CM acknowledge funding from the MRC Centre for Global Infectious Disease Analysis (reference MR/X020258/1), funded by the UK Medical Research Council (MRC). This UK-funded award is carried out in the frame of the Global Health EDCTP3 Joint Undertaking. SB acknowledges support from the Danish National Research Foundation via a chair grant (DNRF160) which also supports NS and MPK. SB acknowledges support from The Eric and Wendy Schmidt Fund For Strategic Innovation via the Schmidt Polymath Award (G-22-63345) which also supports CM and PRL. SB acknowledges support from the Novo Nordisk Foundation via The Novo Nordisk Young Investigator Award (NNF20OC0059309). DAD is supported by a Data Science - Emerging researcher award from Novo Nordisk Fonden (NNF23OC0084647).

Author contributions NS, CM, DAD: study conception and design. NS, RK, PRL, NP: methodology research. NS: practical execution. NS, MPK, DAD: interpretation of results. NS, MPK, CM, DAD, SB: writing of the original draft. All authors contributed to the edition of the original draft.

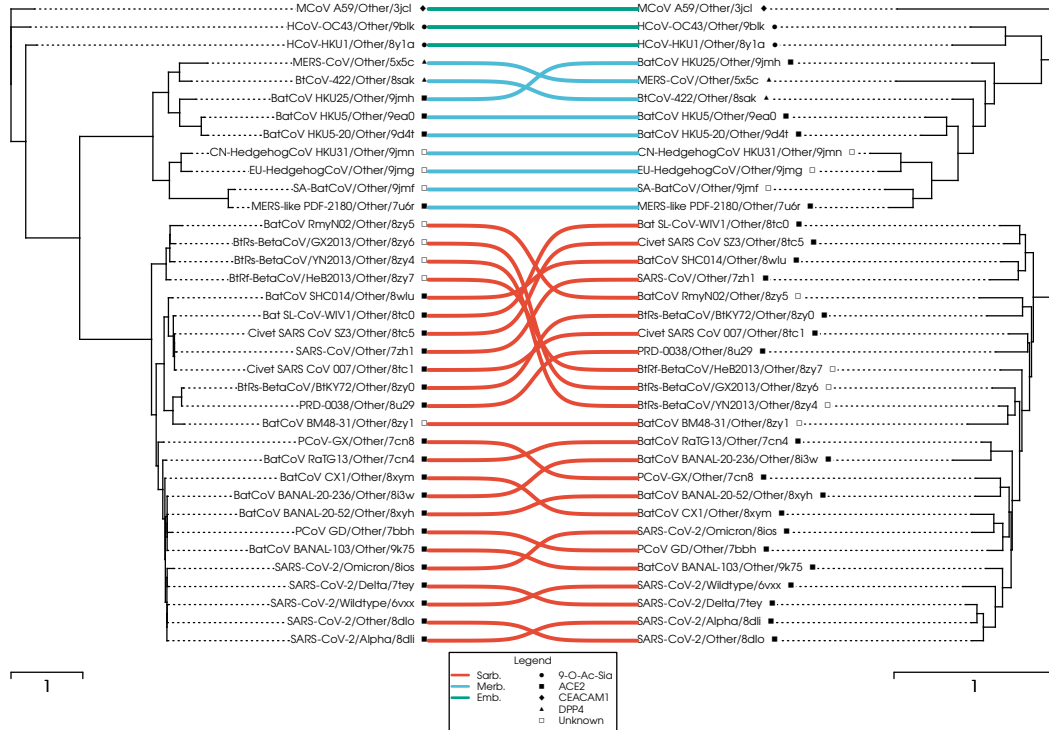
Ethics declaration The authors declare no competing interests.

Rights and permissions This work is licenced under the Creative Commons Attribution 4.0 International Licence (CC BY 4.0). This licence allows others to copy, redistribute, remix, transform, and build upon this work for any purpose, including commercial use, provided appropriate credit is given, a link to the licence is provided, and any changes made are indicated. No additional restrictions beyond those stated in the licence may apply.

Appendix

Supplementary Figures

a RBD: AA vs. 3Di



b NTD: AA vs. 3Di

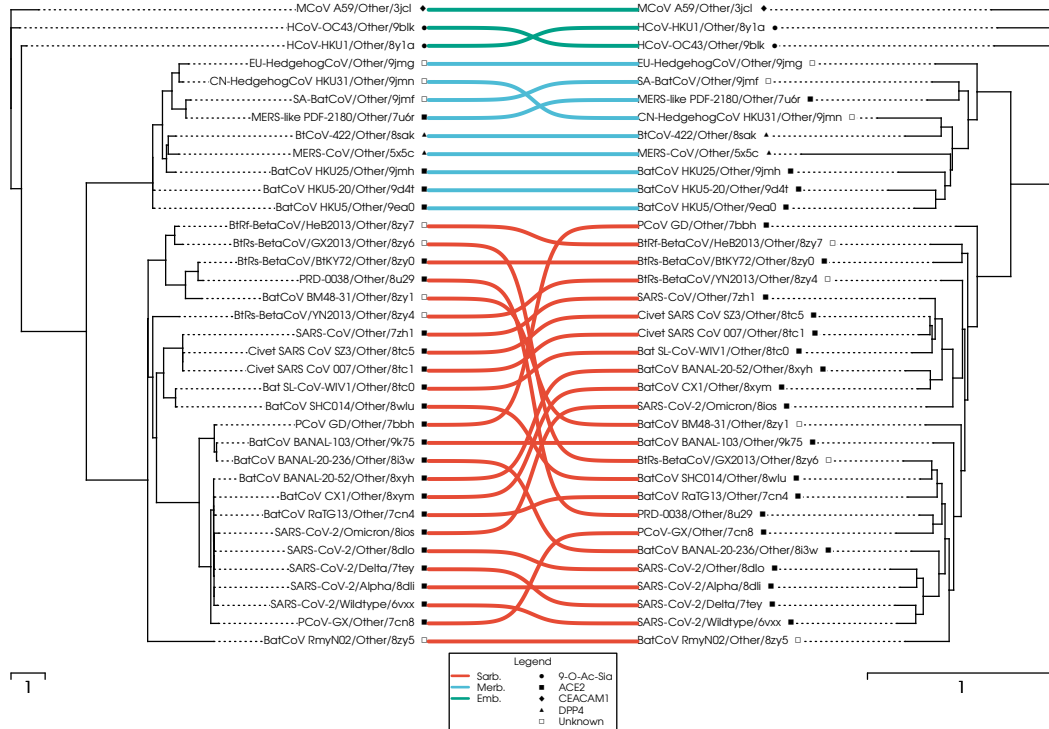
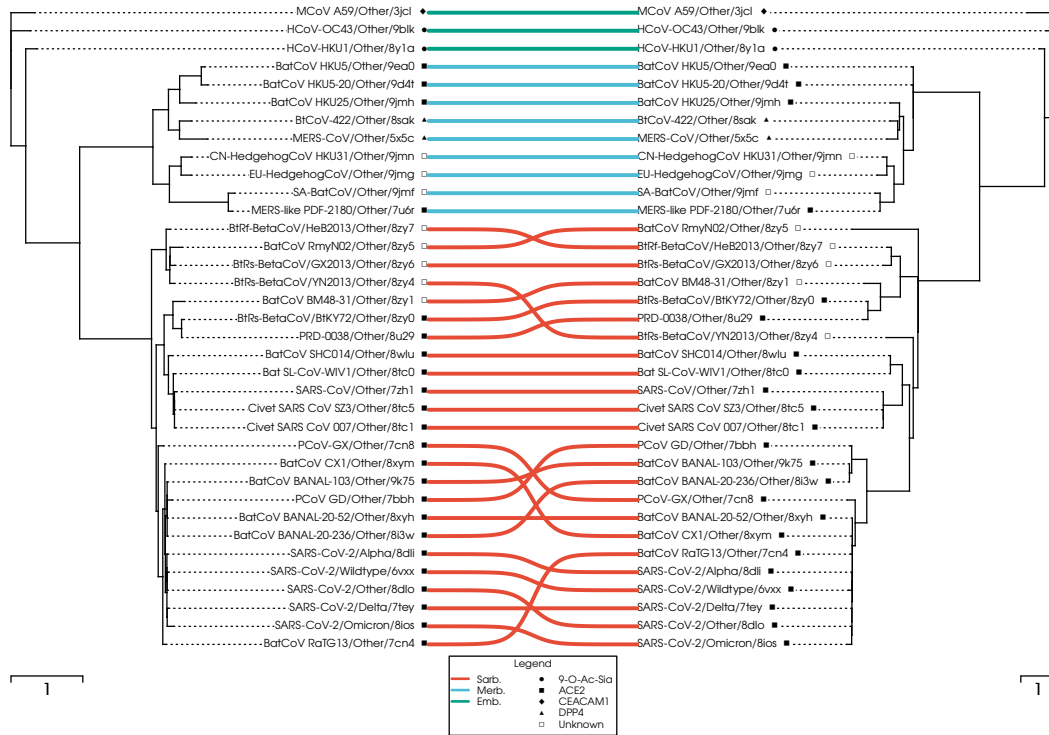
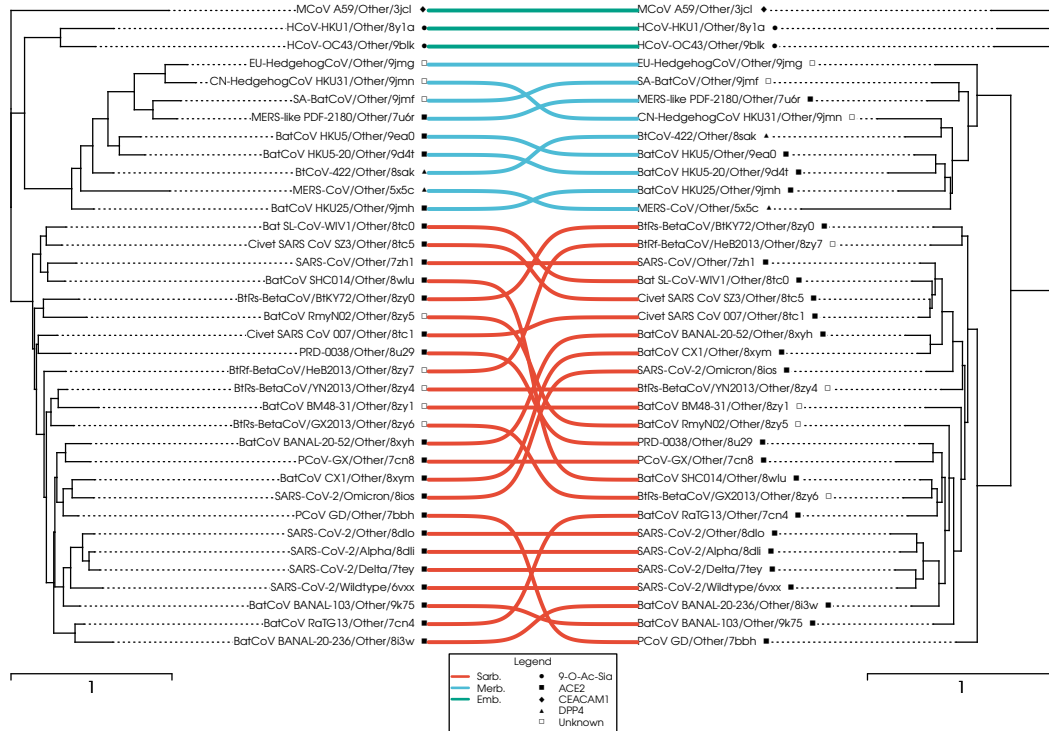


Figure S1. Co-phylogenetic plots of trees from the optimal MAST models (Fig. 3a,3d) covering the RBD (a), NTD ((b)), S2 ((c)) as well as replications of Fig. 4a and 4b with tip labels ((c), (d)).

c AA: RBD vs. NTD**d 3Di: RBD vs. NTD****Figure S1. Co-phylogenetic plots (continued, part 2)**

e S2 (S2 of AA vs. class 3 of 3Di)

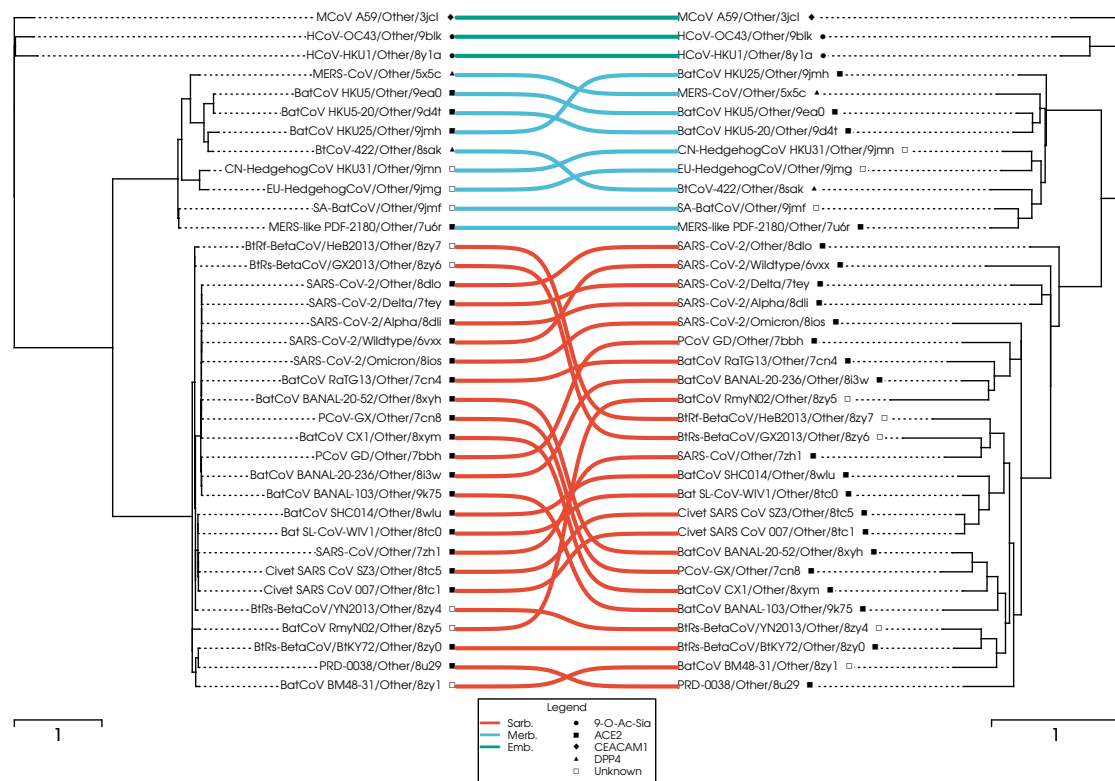


Figure S1. Co-phylogenetic plots (continued, part 3)

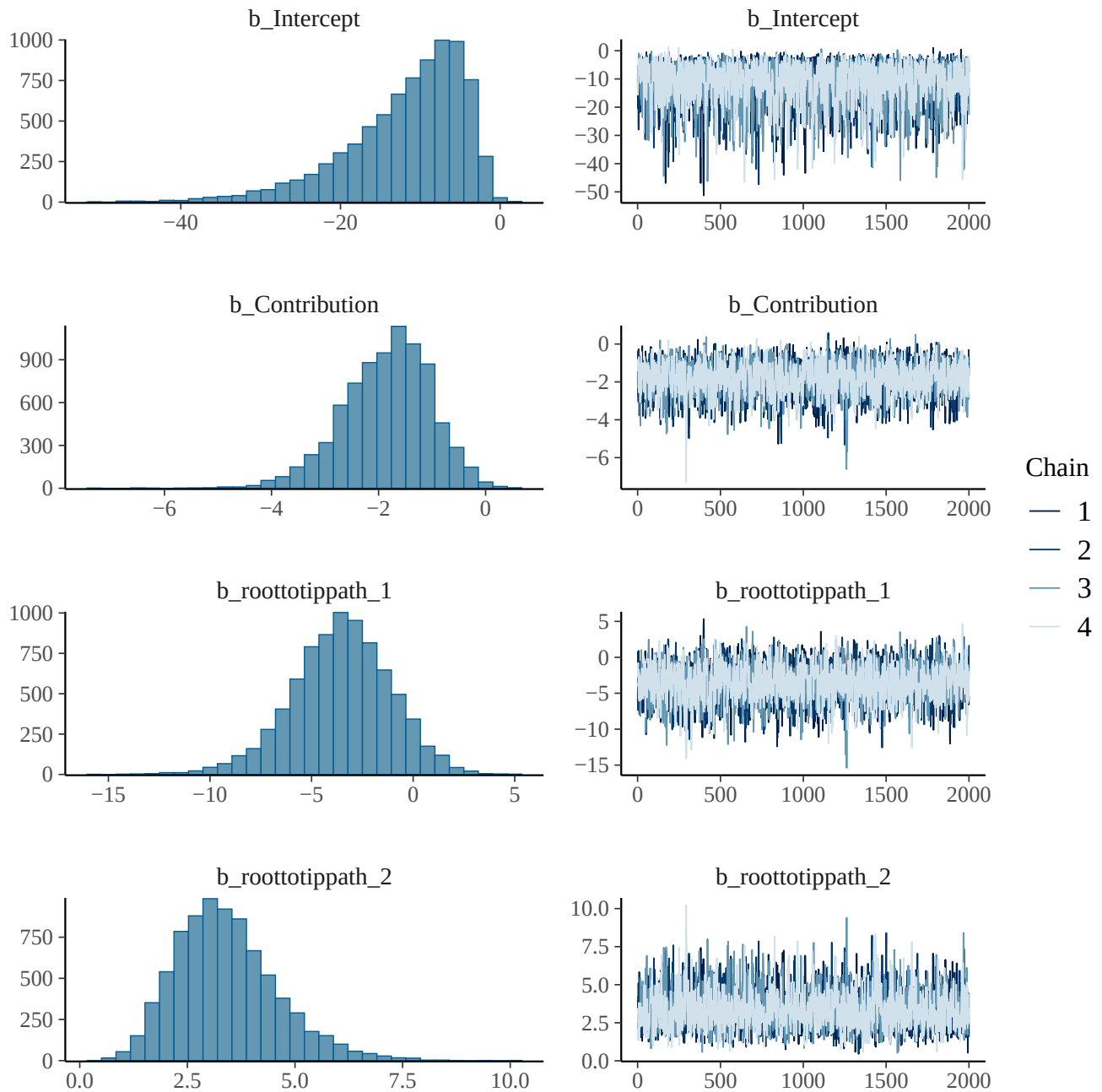


Figure S2. `brms` output for AA, showing density plots for each factor (left) and caterpillar plots (right). Contribution: TDI; `roottotippath_1`: RTT length (RBD); `roottotippath_2`: RTT length (NTD).

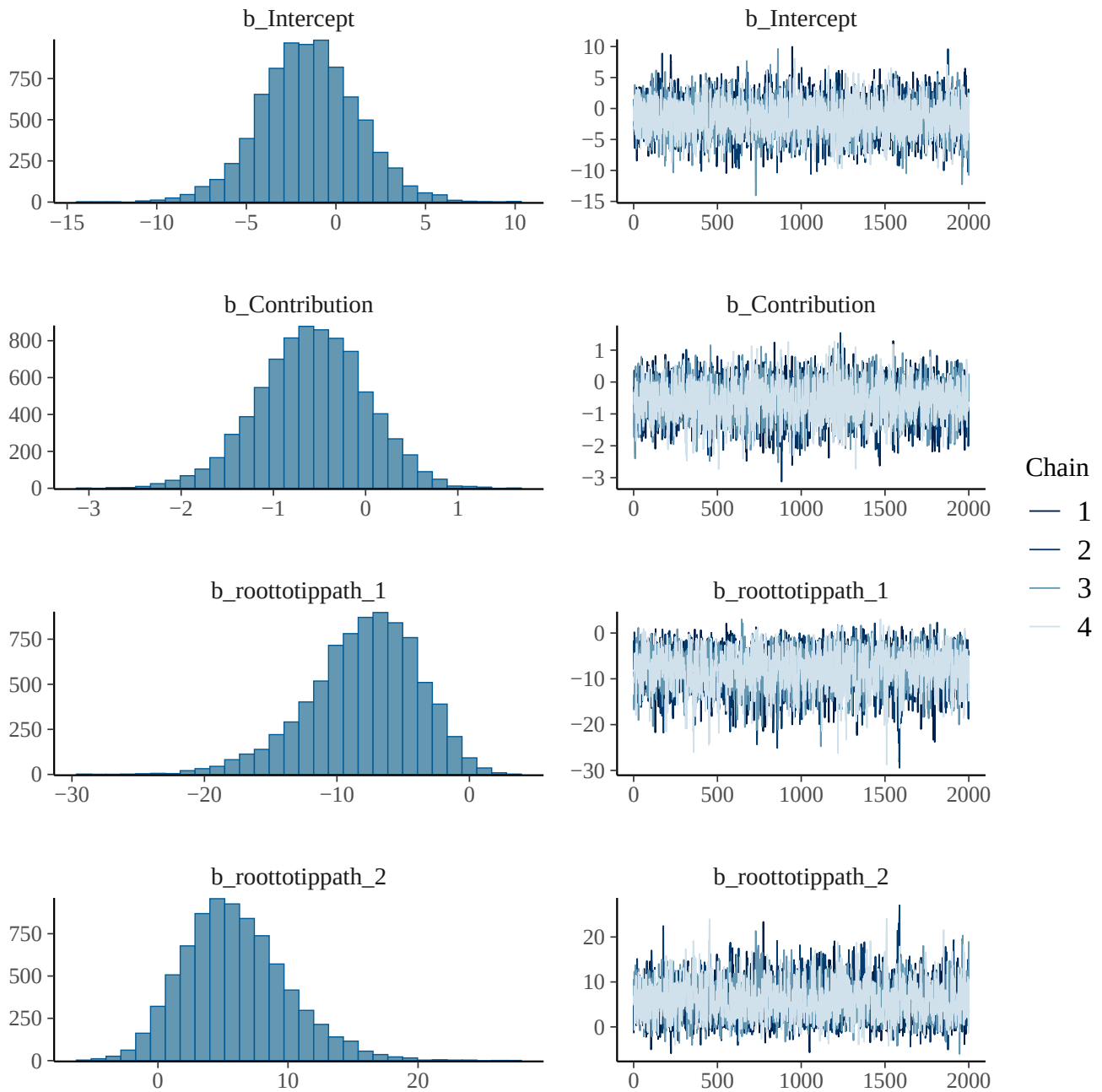


Figure S3. `brms` output for 3Di, showing density plots for each factor (left) and caterpillar plots (right). Contribution: TDI; `roottotippath_1`: RTT length (NTD); `roottotippath_2`: RTT length (RBD).

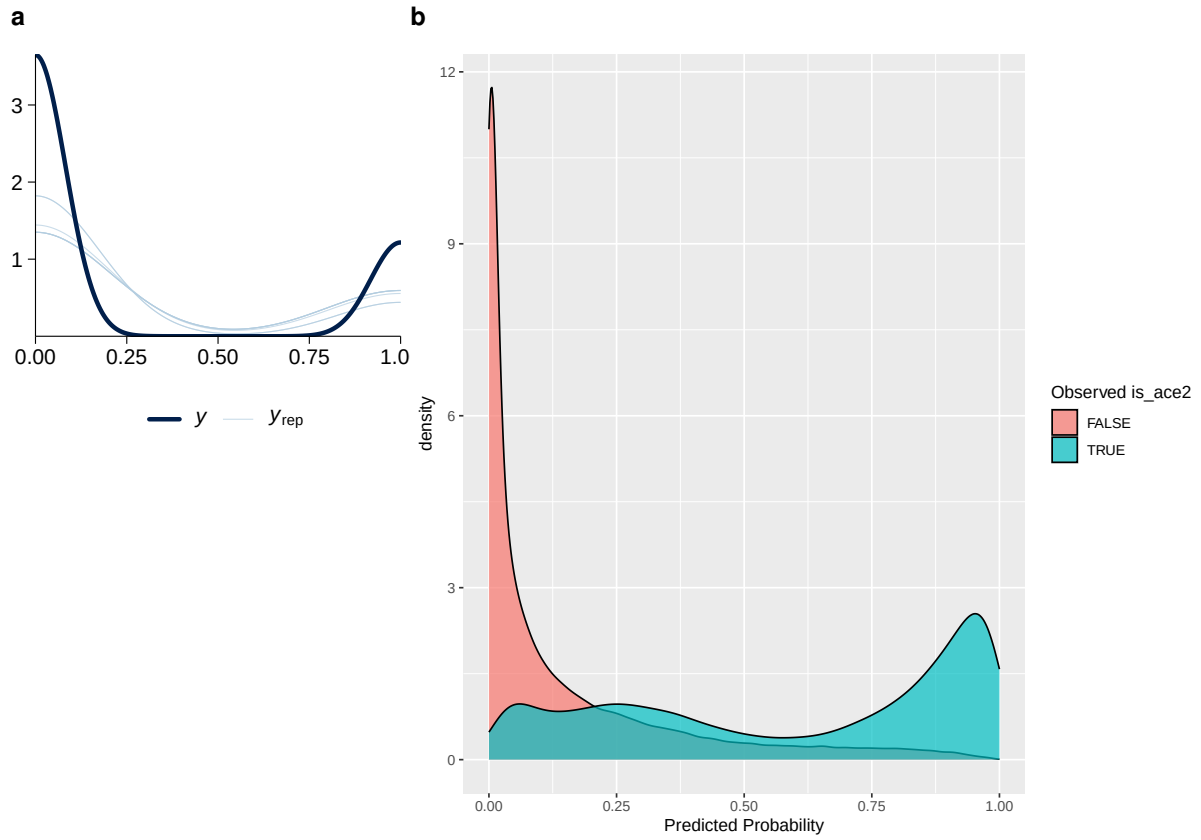


Figure S4. `brms` posterior predictive checks and classification plot for AA.

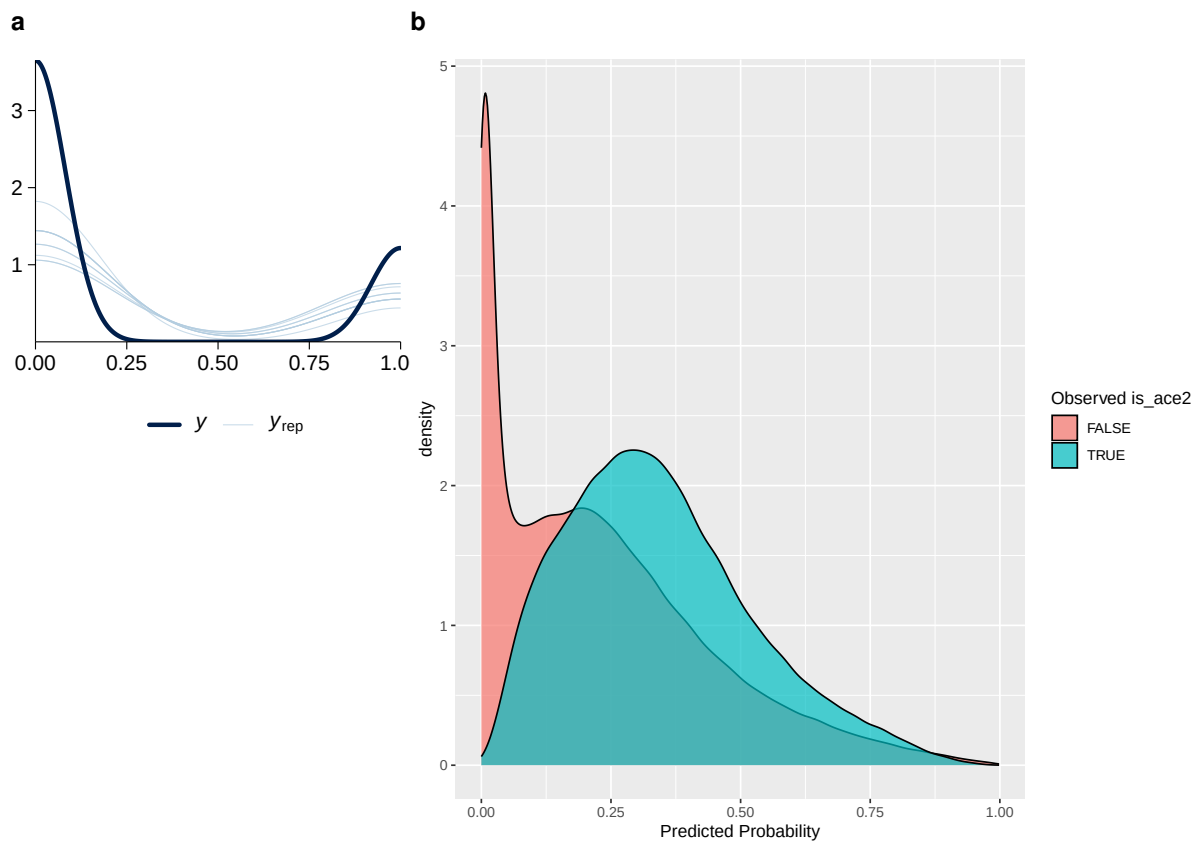


Figure S5. `brms` posterior predictive checks and classification plot for 3Di.

Supplementary Tables

Table S1. ModelFinder output for the ‘subset@3’ AA set (see Fig. 1). Only the top 50 models according to BIC are shown. Bold: best model. Italic: chosen model for subsequent analyses.

Model	LogL	AIC	w-AIC	AICc	w-AICc	BIC	w-BIC
WAG+F+R4	-29354.202	59000.404	8.62E-02	59041.129	1.47E-01	59743.68	9.36E-01
WAG+F+R5	-29349.841	58995.682	9.14E-01	59037.606	8.53E-01	59749.14	6.10E-02
WAG+F+G4	-29377.977	59037.955	-6.05E-10	59075.768	-4.41E-09	59755.77	-2.21E-03
WAG+F+R3	-29368.441	59024.881	-4.17E-07	59064.427	-1.28E-06	59757.97	-7.36E-04
LG+F+R4	-29465.105	59222.21	-5.90E-50	59262.934	-1.00E-49	59965.48	-6.41E-49
LG+F+R5	-29460.718	59217.435	-6.42E-49	59259.359	-6.00E-49	59970.89	-4.29E-50
WAG+F+R2	-29485.649	59255.298	-3.86E-57	59293.683	-2.11E-56	59978.21	-1.11E-51
LG+F+G4	-29490.773	59263.547	-6.23E-59	59301.36	-4.55E-58	59981.37	-2.28E-52
LG+F+R3	-29482.873	59253.746	-8.38E-57	59293.292	-2.57E-56	59986.84	-1.48E-53
WAG+R4	-29545.024	59344.048	-2.06E-76	59374.348	-6.44E-74	59990.59	-2.26E-54
WAG+R5	-29540.714	59339.429	-2.08E-75	59370.745	-3.90E-73	59996.16	-1.40E-55
Q.PFAM+F+R4	-29482.332	59256.664	-1.95E-57	59297.389	-3.31E-57	59999.94	-2.11E-56
WAG+G4	-29570.122	59384.244	-3.85E-85	59412.084	-4.11E-82	60005.34	-1.42E-57
WAG+R3	-29559.566	59369.131	-7.37E-82	59398.433	-3.79E-79	60005.5	-1.31E-57
Q.PFAM+F+R5	-29478.386	59252.772	-1.36E-56	59294.696	-1.27E-56	60006.23	-9.10E-58
Q.PFAM+F+G4	-29507.98	59297.961	-2.10E-66	59335.774	-1.53E-65	60015.78	-7.67E-60
BLOSUM62+F+G4	-29510.06	59302.12	-2.62E-67	59339.933	-1.91E-66	60019.94	-9.59E-61
Q.PFAM+F+R3	-29500.339	59288.677	-2.18E-64	59328.223	-6.68E-64	60021.77	-3.84E-61
BLOSUM62+F+R3	-29500.794	59289.588	-1.38E-64	59329.134	-4.23E-64	60022.68	-2.44E-61
BLOSUM62+F+R4	-29493.747	59279.494	-2.15E-62	59320.219	-3.65E-62	60022.77	-2.33E-61
EX_EHO+R4	-29556.021	59376.042	-2.33E-83	59408.919	-2.00E-81	60048.04	-7.57E-67
EX_EHO+R4	-29556.021	59376.042	-2.33E-83	59408.919	-2.00E-81	60048.04	-7.57E-67
PMB+F+R4	-29506.506	59305.013	-6.17E-68	59345.738	-1.05E-67	60048.29	-6.70E-67
PMB+F+R3	-29516.437	59320.873	-2.22E-71	59360.419	-6.81E-71	60053.96	-3.92E-68
PMB+F+G4	-29527.671	59337.341	-5.90E-75	59375.154	-4.30E-74	60055.16	-2.16E-68
PMB+F+R5	-29505.292	59306.584	-2.82E-68	59348.508	-2.63E-68	60060.04	-1.88E-69
EX_EHO+R5	-29555.698	59379.396	-4.35E-84	59413.336	-2.20E-82	60061.58	-8.71E-70
EX_EHO+R5	-29555.698	59379.396	-4.35E-84	59413.336	-2.20E-82	60061.58	-8.71E-70
Q.PFAM+R4	-29583.041	59420.083	-6.36E-93	59450.383	-1.99E-90	60066.63	-6.97E-71
Q.PFAM+R5	-29578.516	59415.031	-7.95E-92	59446.348	-1.49E-89	60071.76	-5.36E-72
EX_EHO+R3	-29578.287	59416.575	-3.67E-92	59448.407	-5.33E-90	60078.39	-1.94E-73
EX_EHO+R3	-29578.287	59416.575	-3.67E-92	59448.407	-5.33E-90	60078.39	-1.94E-73
EX_EHO+G4	-29590.409	59434.817	-4.02E-96	59465.117	-1.25E-93	60081.36	-4.40E-74
EX_EHO+G4	-29590.409	59434.817	-4.02E-96	59465.117	-1.25E-93	60081.36	-4.40E-74
Q.PFAM+G4	-29608.794	59461.588	-6.18E-102	59489.429	-6.59E-99	60082.68	-2.28E-74
Q.PFAM+R3	-29599.212	59448.424	-4.46E-99	59477.726	-2.29E-96	60084.79	-7.95E-75
LG+R4	-29601.347	59456.693	-7.14E-101	59486.993	-2.23E-98	60103.24	-7.82E-79
LG+R5	-29596.831	59451.663	-8.83E-100	59482.979	-1.66E-97	60108.39	-5.95E-80
LG+G4	-29626.715	59497.429	-1.02E-109	59525.27	-1.09E-106	60118.52	-3.76E-82
LG+R3	-29618.045	59486.089	-2.95E-107	59515.392	-1.52E-104	60122.45	-5.26E-83
JTTDCMUT+F+R4	-29559.456	59410.912	-6.24E-91	59451.636	-1.06E-90	60154.18	-6.77E-90
JTTDCMUT+F+R5	-29555.643	59407.285	-3.82E-90	59449.209	-3.57E-90	60160.74	-2.55E-91
JTTDCMUT+F+G4	-29582.564	59447.128	-8.53E-99	59484.941	-6.22E-98	60164.95	-3.12E-92
JTT+F+R4	-29566.409	59424.818	-5.96E-94	59465.543	-1.01E-93	60168.09	-6.47E-93
JTTDCMUT+F+R3	-29575.174	59438.348	-6.87E-97	59477.893	-2.11E-96	60171.44	-1.21E-93
JTT+F+R5	-29562.74	59421.48	-3.16E-93	59463.404	-2.95E-93	60174.94	-2.11E-94
JTT+F+G4	-29588.628	59459.257	-1.98E-101	59497.07	-1.45E-100	60177.08	-7.25E-95
JTT+F+R3	-29581.271	59450.543	-1.55E-99	59490.088	-4.74E-99	60183.63	-2.73E-96
EX_EHO+R2	-29652.272	59560.543	-2.01E-123	59591.349	-4.87E-121	60212.18	-1.73E-102
EX_EHO+R2	-29652.272	59560.543	-2.01E-123	59591.349	-4.87E-121	60212.18	-1.73E-102

Table S2. ModelFinder output for the ‘subset@3’ 3Di set (see Fig. 1). Values displayed as 0 are smaller than the smallest positive representable double-precision number ($\approx -1.798+308$). Bold: best model. Italic: chosen model for subsequent analyses.

Model	LogL	AIC	w-AIC	AICc	w-AICc	BIC	w-BIC
<i>Q.3DI.AF+F+R4</i>	-57694.1	<i>115680.1</i>	<i>7.81E-01</i>	<i>115729.7</i>	<i>8.82E-01</i>	<i>116398.6</i>	<i>9.98E-01</i>
Q.3DI.AF+F+R5	-57693.3	115682.7	2.19E-01	115733.7	1.18E-01	116410.9	-2.05E-03
Q.3DI.AF+F+G4	-57732	115746	-3.88E-15	115792	-2.64E-14	116439.8	-1.09E-09
Q.3DI.AF+F+R3	-57738.5	115765.1	-2.83E-19	115813.2	-6.62E-19	116473.6	-4.96E-17
Q.3DI.AF+R4	-57867.4	115988.8	-7.45E-68	116025.5	-5.14E-65	116613.7	-1.91E-47
Q.3DI.AF+R5	-57862.9	115983.7	-9.30E-67	116021.7	-3.44E-64	116618.5	-1.74E-48
Q.3DI.AF+G4	-57910.8	116065.7	-1.49E-84	116099.4	-4.65E-81	116666	-8.41E-59
Q.3DI.AF+R3	-57915.6	116081.3	-6.17E-88	116116.8	-7.84E-85	116696.4	-2.16E-65
Q.3DI.AF+F+R2	-57958.1	116200.3	-8.90E-114	116246.9	-4.26E-113	116899	-2.14E-109
Q.3DI.LLM+F+R4	-58064.7	116421.3	-8.92E-162	116470.9	-1.01E-161	117139.7	-1.14E-161
Q.3DI.LLM+F+R5	-58064.3	116424.6	-1.69E-162	116475.7	-9.09E-163	117152.9	-1.57E-164
Q.3DI.LLM+F+G4	-58099.2	116480.3	-1.35E-174	116526.3	-9.21E-174	117174.2	-3.81E-169
Q.3DI.AF+R2	-58170.4	116586.7	-1.09E-197	116621	-2.52E-194	117191.9	-5.23E-173
Q.3DI.LLM+F+R3	-58108.8	116505.6	-4.54E-180	116553.7	-1.06E-179	117214.1	-7.96E-178
Q.3DI.LLM+F+R2	-58324.8	116933.7	-4.88E-273	116980.4	-2.33E-272	117632.4	-1.17E-268
Q.3DI.LLM+R4	-58634.9	117523.9	0.00E+00	117560.6	0.00E+00	118148.8	0.00E+00
Q.3DI.LLM+R5	-58630	117518	0.00E+00	117556	0.00E+00	118152.8	0.00E+00
Q.3DI.LLM+G4	-58680	117604	0.00E+00	117637.7	0.00E+00	118204.3	0.00E+00
Q.3DI.LLM+R3	-58681.5	117612.9	0.00E+00	117648.4	0.00E+00	118228	0.00E+00
Q.3DI.LLM+R2	-58935.9	118117.9	0.00E+00	118152.2	0.00E+00	118723.1	0.00E+00
Q.3DI.AF+F	-59323.4	118926.8	0.00E+00	118972.1	0.00E+00	119615.7	0.00E+00
Q.3DI.AF	-59572.3	119386.6	0.00E+00	119419.8	0.00E+00	119982	0.00E+00
Q.3DI.LLM+F	-59728.4	119736.7	0.00E+00	119782	0.00E+00	120425.6	0.00E+00
Q.3DI.LLM	-60342.6	120927.2	0.00E+00	120960.4	0.00E+00	121522.6	0.00E+00
VK23+R4	-60788.2	121830.4	0.00E+00	121867.2	0.00E+00	122455.4	0.00E+00
VK23+R5	-60788.2	121834.4	0.00E+00	121872.3	0.00E+00	122469.1	0.00E+00
VK23+G4	-60840.8	121925.7	0.00E+00	121959.4	0.00E+00	122526	0.00E+00
VK23+R3	-60837.6	121925.2	0.00E+00	121960.7	0.00E+00	122540.3	0.00E+00
VK23+F+R5	-60790.5	121877	0.00E+00	121928	0.00E+00	122605.2	0.00E+00
VK23+F+R6	-60789.1	121878.3	0.00E+00	121930.8	0.00E+00	122616.4	0.00E+00
VK23+F+R4	-60809.4	121910.7	0.00E+00	121960.3	0.00E+00	122629.1	0.00E+00
VK23+F+G4	-60844.5	121971	0.00E+00	122017	0.00E+00	122664.8	0.00E+00
VK23+F+R3	-60854	121996.1	0.00E+00	122044.2	0.00E+00	122704.7	0.00E+00
VK23+R2	-61113.1	122472.1	0.00E+00	122506.4	0.00E+00	123077.3	0.00E+00
VK23+F+R2	-61124.6	122533.2	0.00E+00	122579.9	0.00E+00	123231.9	0.00E+00
VK23+F	-62723.4	125726.9	0.00E+00	125772.1	0.00E+00	126415.8	0.00E+00
VK23	-62802.1	125846.2	0.00E+00	125879.3	0.00E+00	126441.6	0.00E+00

Table S3. `brms` model summary (AA). Significant differences (determined if the credible intervals exclude zero) with respect to the negative control are shown in bold. CrI: credible interval (l: lower, u: upper). $\hat{R}$: Gelman-Rubin statistic for Markov chain Monte Carlo convergence; convergence is deemed suitable when $\hat{R}$ is close to 1. ESS: effective sample size.

	Estimate	Est.Error	l-95% CrI	u-95% CrI	$\hat{R}$	Bulk ESS	Tail ESS
Intercept	-11.852	7.497	-30.684	-2.238	1.001	2308.328	2785.365
TDI	-1.847	0.823	-3.608	-0.395	1.002	2602.384	2794.829
RTT length (RBD)	-3.553	2.432	-8.616	1.041	1.001	2506.952	2976.58
RTT length (NTD)	3.409	1.181	1.481	6.117	1.002	2051.326	2753.036

Table S4. `brms` model summary (3Di). Significant differences (determined if the credible intervals exclude zero) with respect to the negative control are shown in bold. CrI: credible interval (l: lower, u: upper). $\hat{R}$: Gelman-Rubin statistic for Markov chain Monte Carlo convergence; convergence is deemed suitable when $\hat{R}$ is close to 1. ESS: effective sample size.

	Estimate	Est.Error	l-95% CrI	u-95% CrI	$\hat{R}$	Bulk ESS	Tail ESS
Intercept	-1.503	2.712	-6.930	3.843	1.001	4271.834	4322.361
TDI	-0.579	0.582	-1.761	0.521	1.002	3523.126	3851.029
RTT length (RBD)	-7.980	4.160	-17.356	-0.969	1.001	2620.180	2606.749
RTT length (NTD)	5.961	4.013	-0.988	14.798	1.002	2699.201	2285.781